

Design and Analysis of Bird Wing Morphing Aircraft.

Guided by: Prof: Y.C. Arun {Assistant professor}

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Introduction

Birds reshape their wings continuously to optimize lift, drag, stability and energy efficiency across diverse flight condition This biological principle has inspired a new generation of Inspired Drones (AIDs).

Objectives

- To study the kinematics and aerodynamics of bird flight.
- To design a mechanical system capable of replicating wing morphing (e.g., span retraction, sweeping, or twisting).
- To fabricate a working desktop or RC model to demonstrate th mechanism.
- To analyze the feasibility and performance of the morphin mechanism.

Why do we need Bio Inspired Morphing wings?

- **The Problem with Conventional Wings:** Traditional aircraft use rigid wings with hinged flaps (ailerons, elevators), which create drag and are less efficient.

- **The Bio-Inspired Solution:** Birds alter their wingspan, sweep, and twist to optimize flight efficiency across different speeds and maneuvers.



Inspired By :

PEREGRINE FALCON



Bird Wing Morphing Aircraft.

06/04/2026

PEREGRINE FALCON

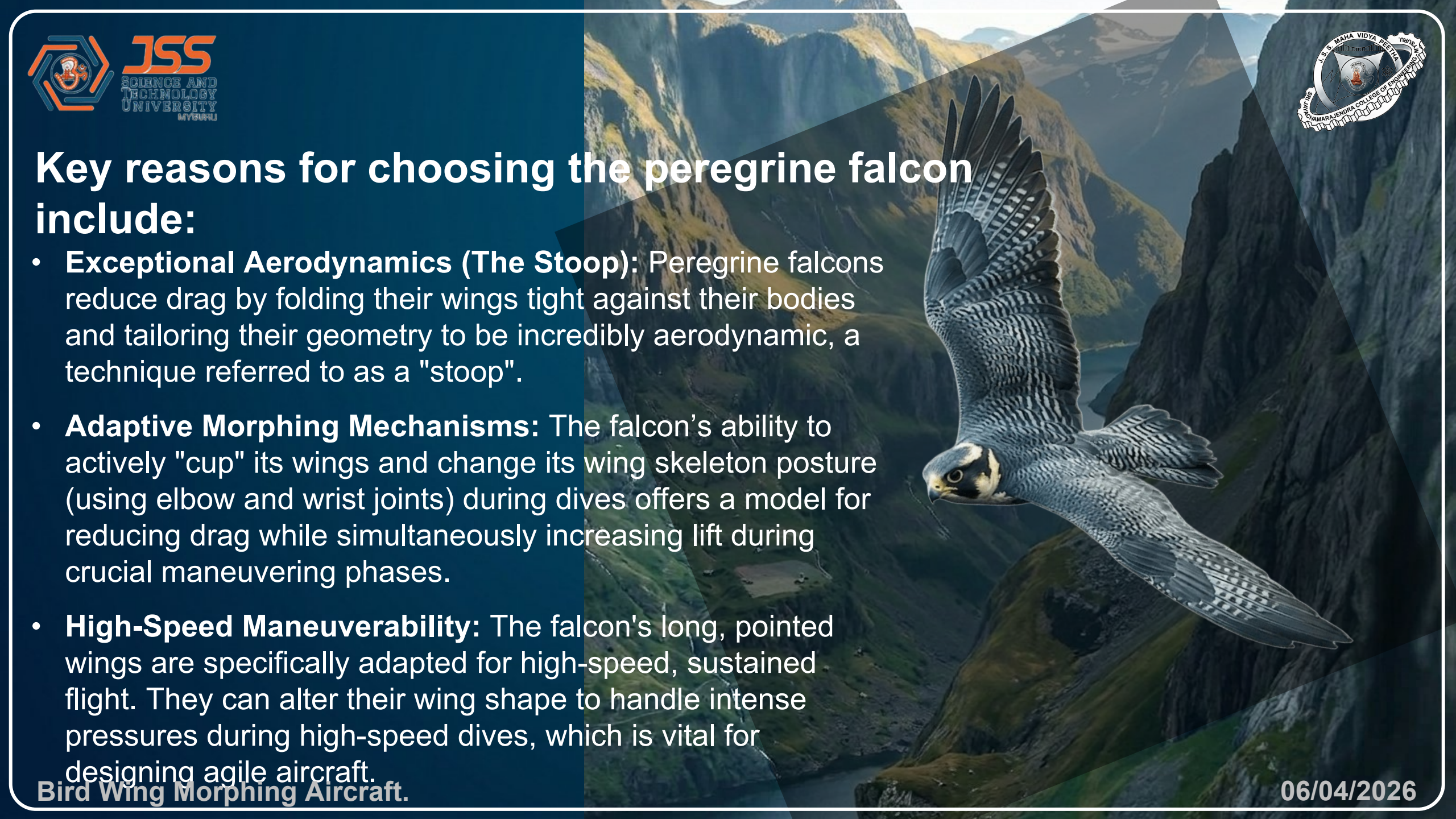
The peregrine falcon (*Falco peregrinus*) is a globally distributed bird of prey, famous for being the fastest animal on earth. During its hunting dive (stoop), it can exceed speeds of 200 mph (320 km/h). These birds are found on every continent except Antarctica and have adapted to live in both remote cliffs and urban

Why to consider Peregrine Falcon?

The peregrine falcon was considered for wing-morphing plane technology because it is the fastest animal on Earth, capable of reaching speeds over 200 mph (320+ km/h) by actively changing its wing shape to maximize aerodynamic efficiency.

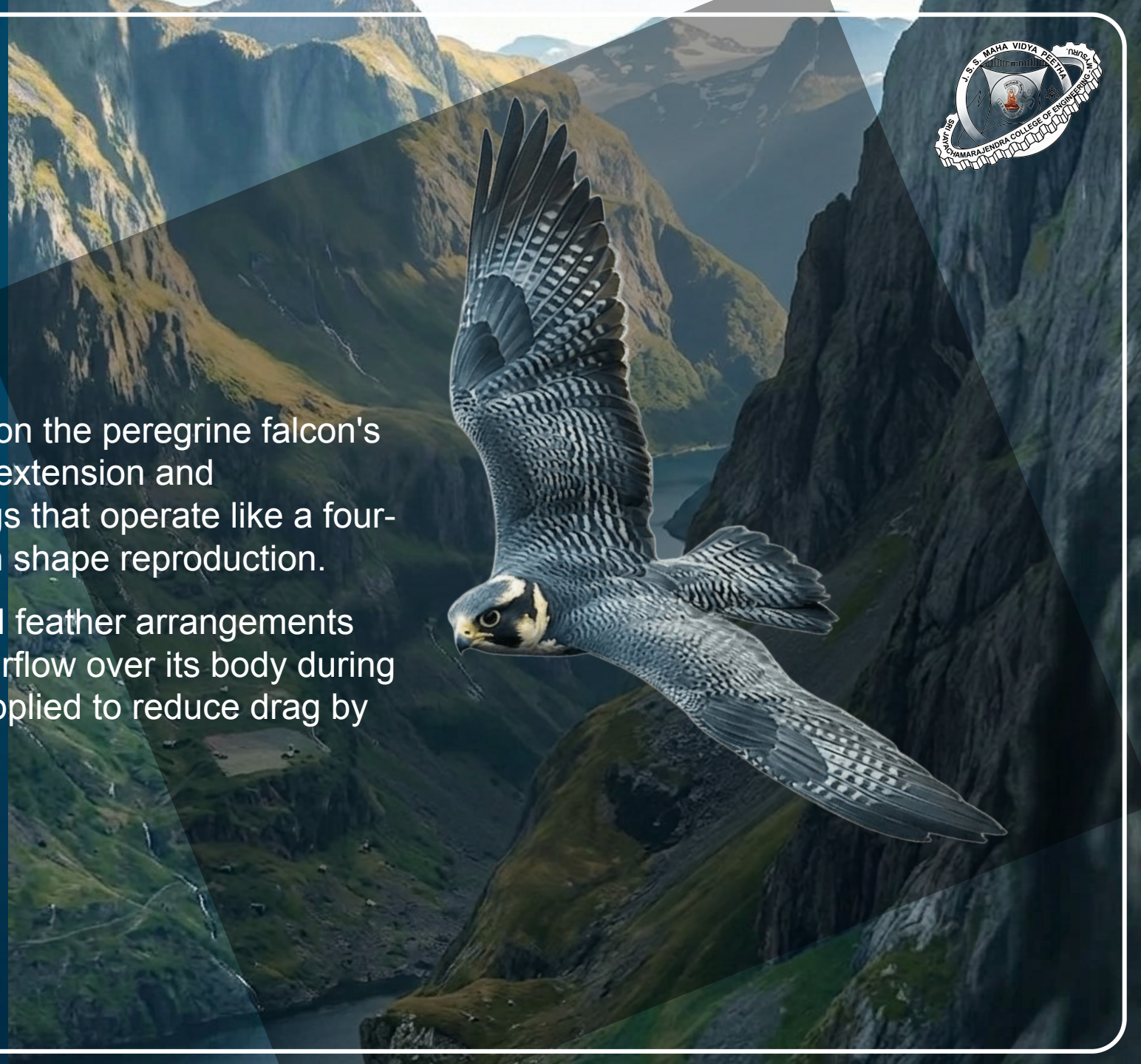
Key reasons for choosing the peregrine falcon include:

- **Exceptional Aerodynamics (The Stoop):** Peregrine falcons reduce drag by folding their wings tight against their bodies and tailoring their geometry to be incredibly aerodynamic, a technique referred to as a "stoop".
- **Adaptive Morphing Mechanisms:** The falcon’s ability to actively "cup" its wings and change its wing skeleton posture (using elbow and wrist joints) during dives offers a model for reducing drag while simultaneously increasing lift during crucial maneuvering phases.
- **High-Speed Maneuverability:** The falcon's long, pointed wings are specifically adapted for high-speed, sustained flight. They can alter their wing shape to handle intense pressures during high-speed dives, which is vital for designing agile aircraft.



designing agile aircraft.

- **Bio-inspired Structural Design:** Studies on the peregrine falcon's skeletal structure (bone movement during extension and contraction) are used to design bionic wings that operate like a four-bar mechanism, allowing for high-precision shape reproduction.
- **Flow Control:** The falcon uses specialized feather arrangements (like coverts) and wing shapes to control airflow over its body during high-speed flight, a principle that can be applied to reduce drag by 30% and increase lift by 45% in aircraft.



Abstract:

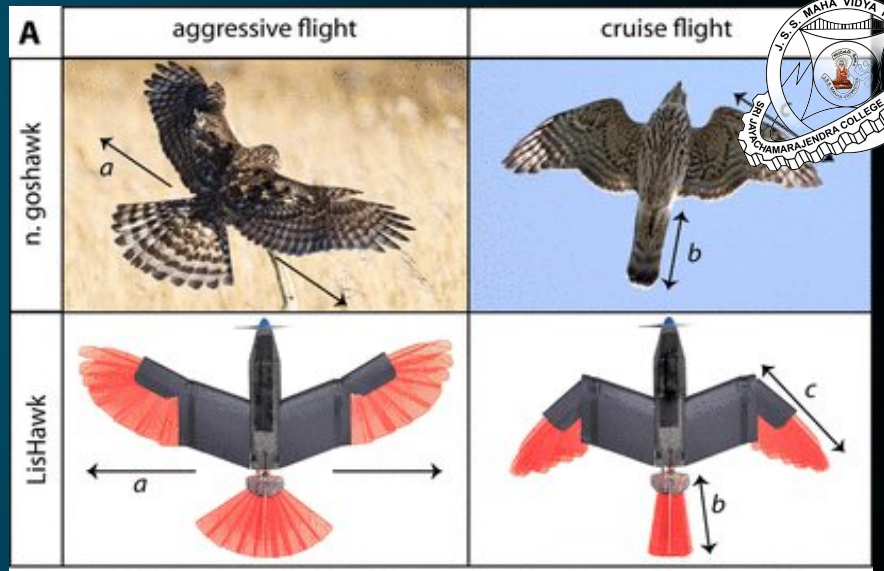
Birds achieve remarkable aerodynamic efficiency through the continuous adaptation of their wing shape during flight — altering camber, span, and twist in real time to respond to changing flight conditions. Inspired by this biological mechanism, this project presents the design, aerodynamic simulation, fabrication, and performance evaluation of a bird wing morphing aircraft. The proposed system uses flexible smart skin actuated by Shape Memory Alloy (SMA) wires and servo-driven ribs to achieve variable camber morphing, mimicking leading-edge and trailing-edge deformation observed in avian wings. A baseline airfoil (NACA 2412) was selected, and CAD models of the morphing wing were developed. The morphed wing can achieve up to 23% improvement in the lift-to-drag ratio compared to the rigid baseline, with significant delay in stall onset. These findings confirm that bird-inspired morphing wings offer superior aerodynamic adaptability for next-generation UAVs, MAVs, and sustainable aviation platforms.

Key Avian-Inspired Drone Systems

LisHawk (EPFL, 2023): 1.1 m wingspan, 460 g; feathered morphing wings & tail; tail-twist enables banked turns

The **LisHawk**, developed by the **Laboratory of Intelligent Systems (LIS) at EPFL in 2023**, is a 1.1 m wingspan, 460 g, bio-inspired drone designed to mimic the **northern goshawk's agility**. It features feathered morphing wings and a tail capable of changing shape in flight to switch between efficient cruising and agile, aggressive flight.

Experimental Setup: Indoor wind tunnel + 26-camera Opti Track motion capture at 240 Hz
Validation: Physical perturbation tests, turbulence tests, actuator failure mid-flight, Bayesian energy optimization



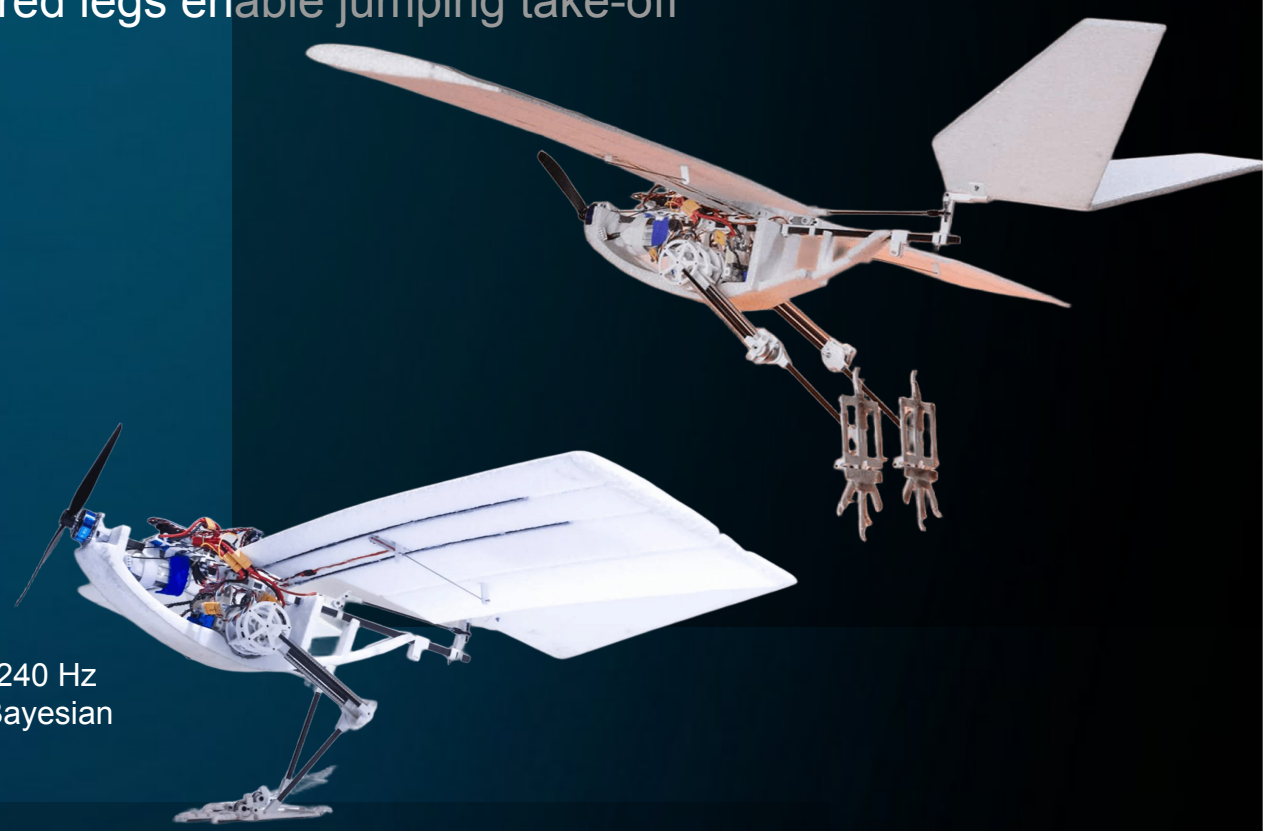
	northern goshawk	LisHawk
wing span (*a*) (m)	0.89 - 1.22	1.05
tail length (*b*) (m)	0.20 - 0.27	0.24
wing chord (*c*) (m)	0.28 - 0.35	0.32



Key Avian-Inspired Drone Systems

RAVEN (EPFL, 2024): 1.0 m wingspan, 620 g; bird-inspired legs enable jumping take-off (2.4 m/s in 0.17 s); walks, hops, jumps

RAVEN (Robotic Avian-inspired Vehicle for multiple Environments), developed by researchers at **EPFL's Laboratory of Intelligent Systems (LIS)** in 2024, is a 620-gram, **1.0-meter wingspan bird-inspired drone** designed to traverse complex environments. It features multifunctional, **bio-inspired legs with torsion springs that allow it to walk, hop, and perform a vertical jumping takeoff to enter flight.**



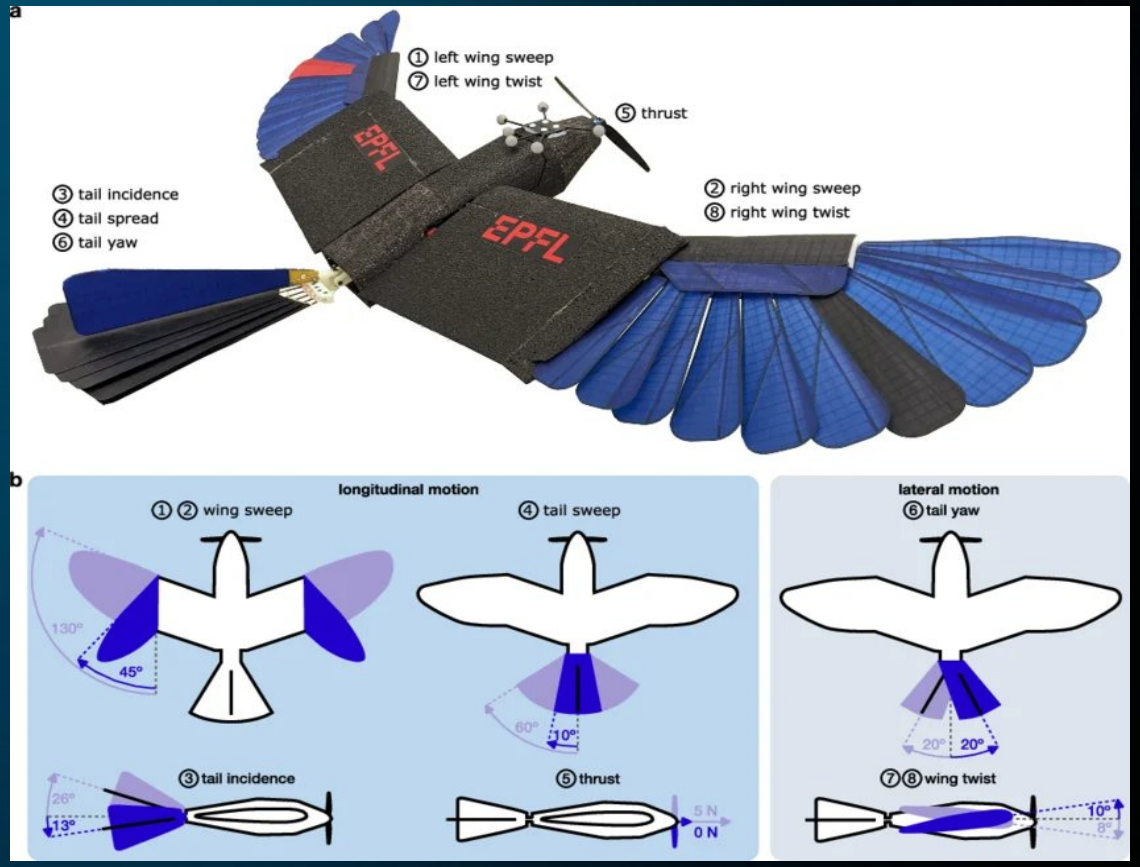
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Key Avian-Inspired Drone Systems

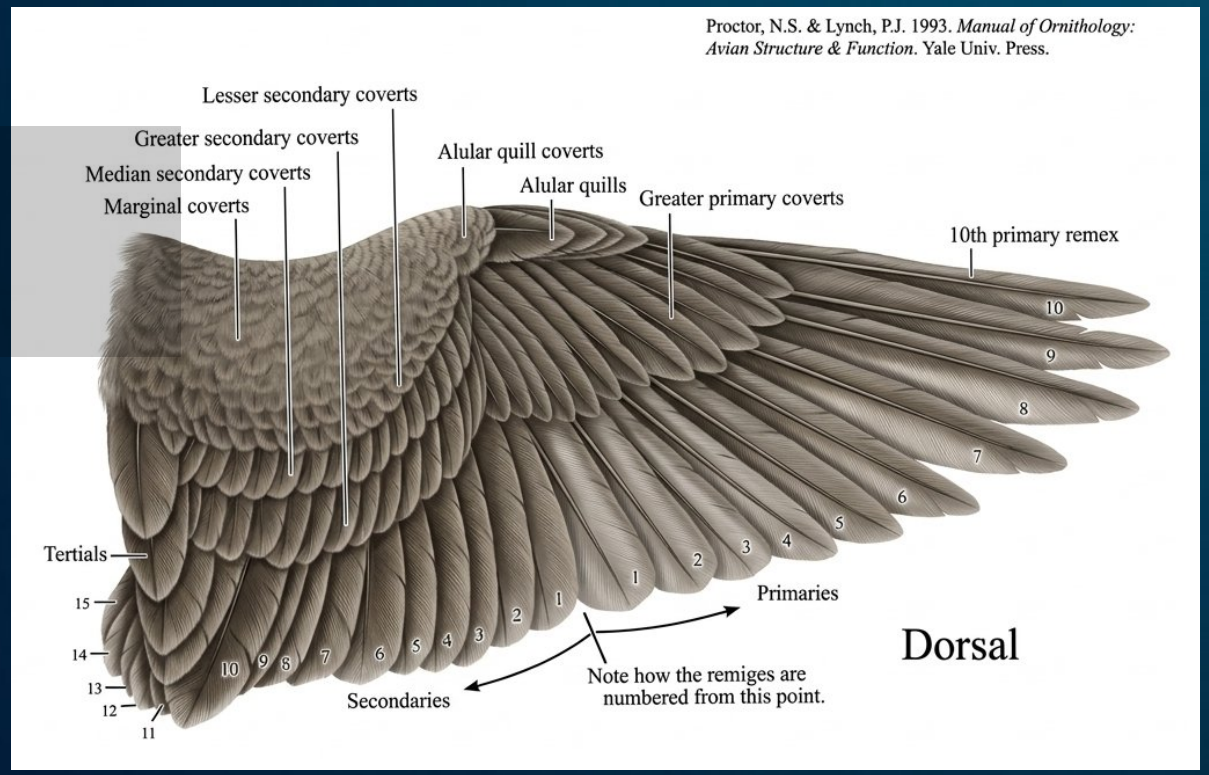
LisEagle (EPFL, 2024): 1.52 m wingspan, 752 g, 8 DoF; stable flight in turbulence; 11.5% energy saving via morphing

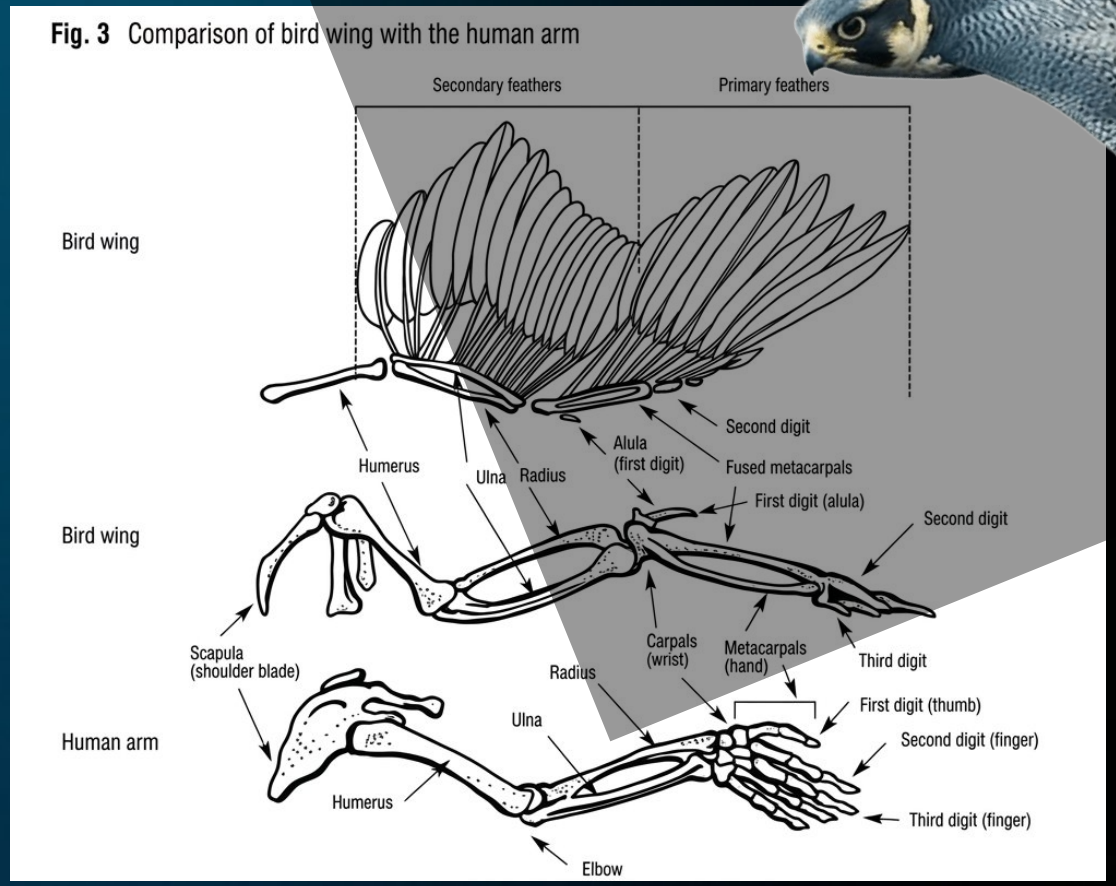
LisEagle is an avian-inspired drone developed by the **Laboratory of Intelligent Systems (LIS) at EPFL in 2024**, designed to replicate the flight characteristics of birds of prey using feathered morphing wings and a tail. The drone features **8 degrees of freedom (DoF)**, a **1.52 m wingspan**, and a ready-to-fly mass of approximately **752 g**.

Experimental Setup: Indoor wind tunnel + 26-camera Opti Track motion capture at 240 Hz
Validation: Physical perturbation tests, turbulence tests, actuator failure mid-flight, Bayesian energy optimization



To Begin, Let us examine bird Anatomy and the Mechanisms they utilize for highly Maneuverable Flight…





To Begin, Let us examine bird Anatomy and the Mechanisms they utilize for highly Maneuverable Flight…



Cruising



Diving



Landing

DESIGN CONCEPT

1. **Bird Wing vs Morphing Aircraft Wing**

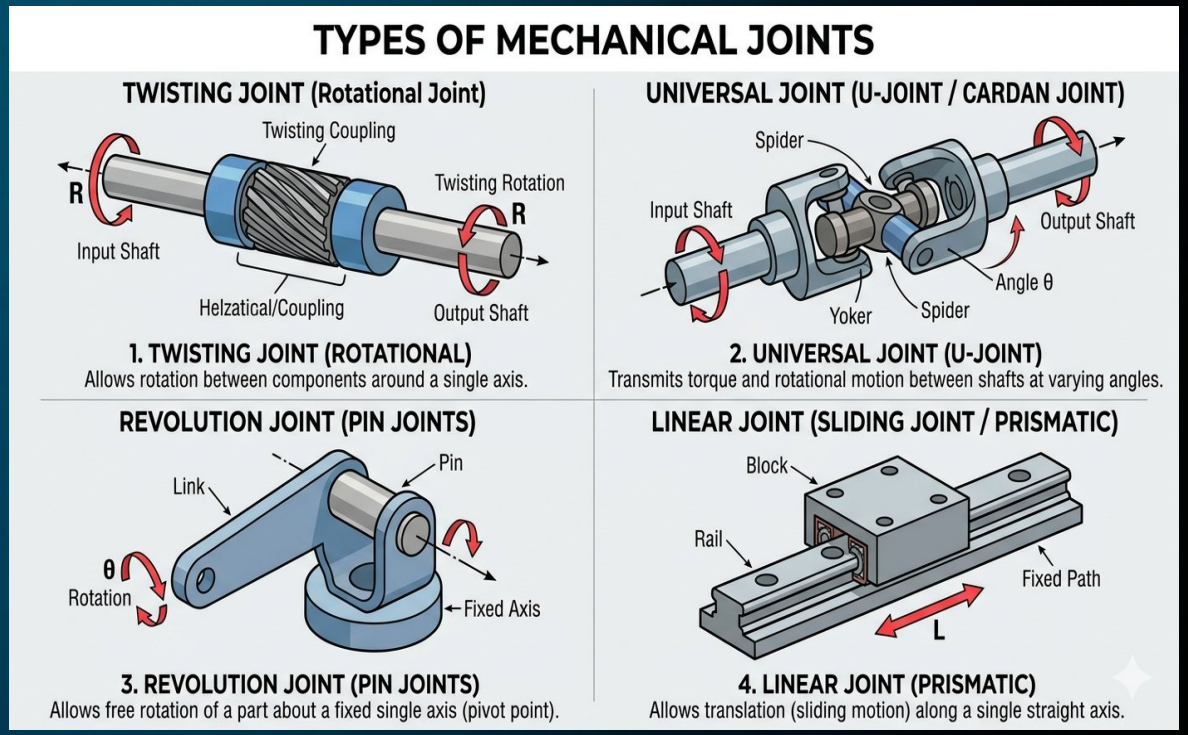
The fundamental inspiration is the avian wing, which exhibits continuous structural compliance and actuation. Figure 1 illustrates the comparison between a bird wing anatomy and the proposed morphing aircraft wing design, highlighting how the flexible, adaptive structure of a bird's wing is translated into an engineered morphing system.

2. **Airfoil Cross-Section — Baseline vs Morphed**

The baseline airfoil **NACA 2412** provides a good balance of lift and stability at low Reynolds numbers. During high-lift phases (landing, tight turns), the camber is increased to NACA 6412 via SMA-actuated trailing edge deflection. Figure 2 shows both profiles overlaid for comparison.

Types of Joints that can be used.

1. Twisting Joint : At the wings where the body and the airfoil joins
2. Liner joints: In the wings and tails
3. Universal joints: for the tail mechanism
4. Revolute Joint (Pin Joint): At the servo horns and primary wing pivots

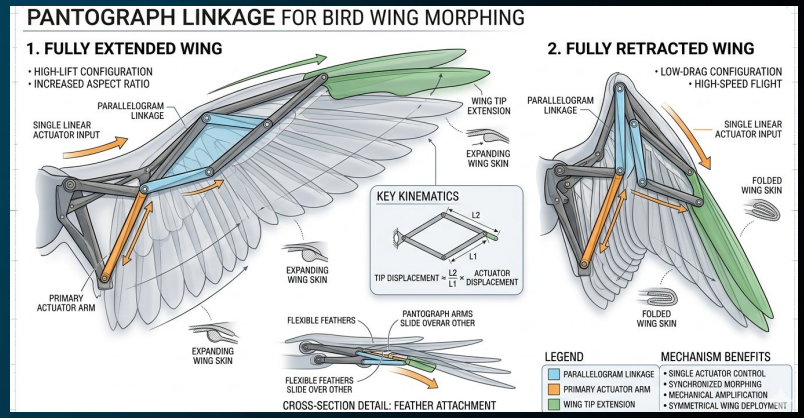


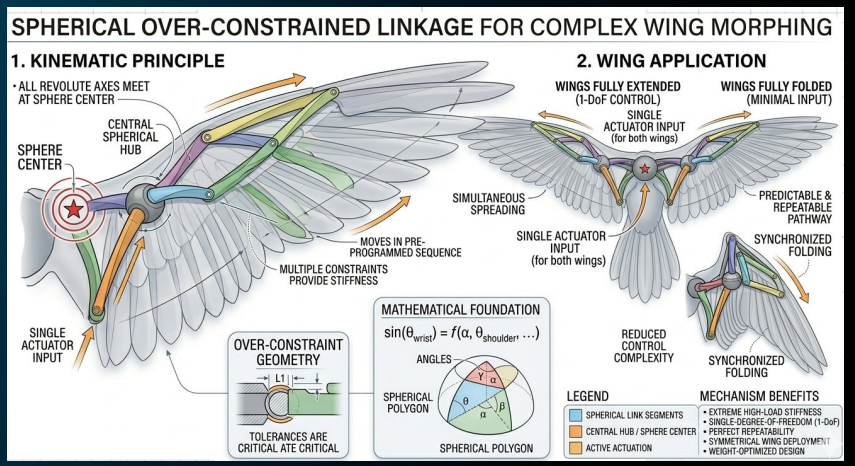
Mechanism (Links) that we Can Use for Achieving the flight in Air.

Mechanisms that Can be Used for Wings

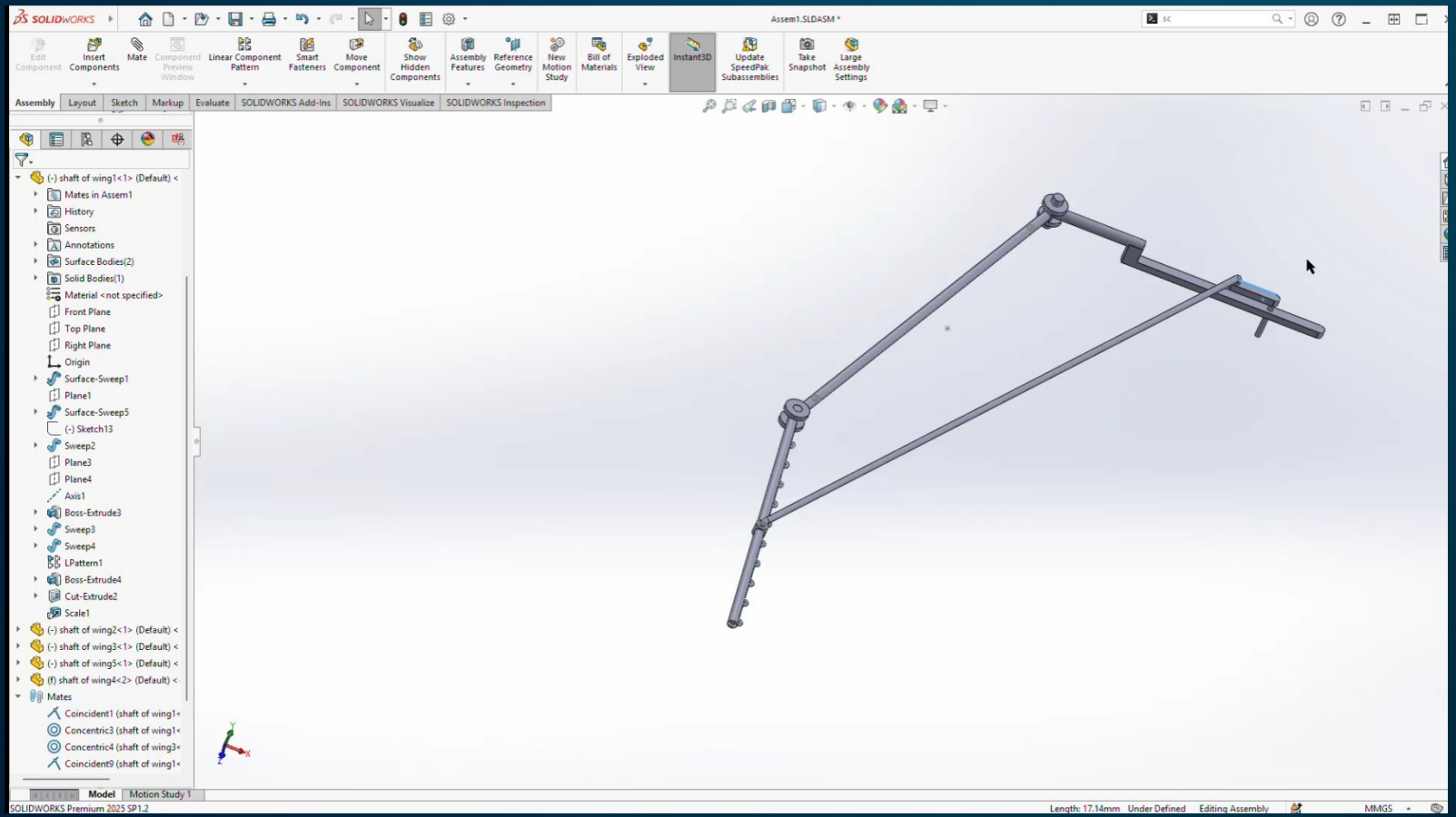
To replicate the bone structure (the humerus, radius, and ulna) of a bird wing, you need linkages that can fold and expand smoothly.

- **Pantograph Linkages:** These are excellent for achieving synchronized, proportional extension. As the main actuator pushes the "shoulder," the rest of the wing scales out in a beautiful, linear motion.
- **Four-Bar Mechanisms:** The bedrock of mechanical engineering. By calculating the lengths of a non-parallel four-bar linkage, you can make the outer "hand" of the wing sweep backward and twist slightly as the inner wing extends.
- **Spherical Over-constrained Linkages:** If you need your model to move in three dimensions simultaneously (like flapping combined with pitching), spherical linkages allow rotation around a single central point, mimicking a shoulder joint.





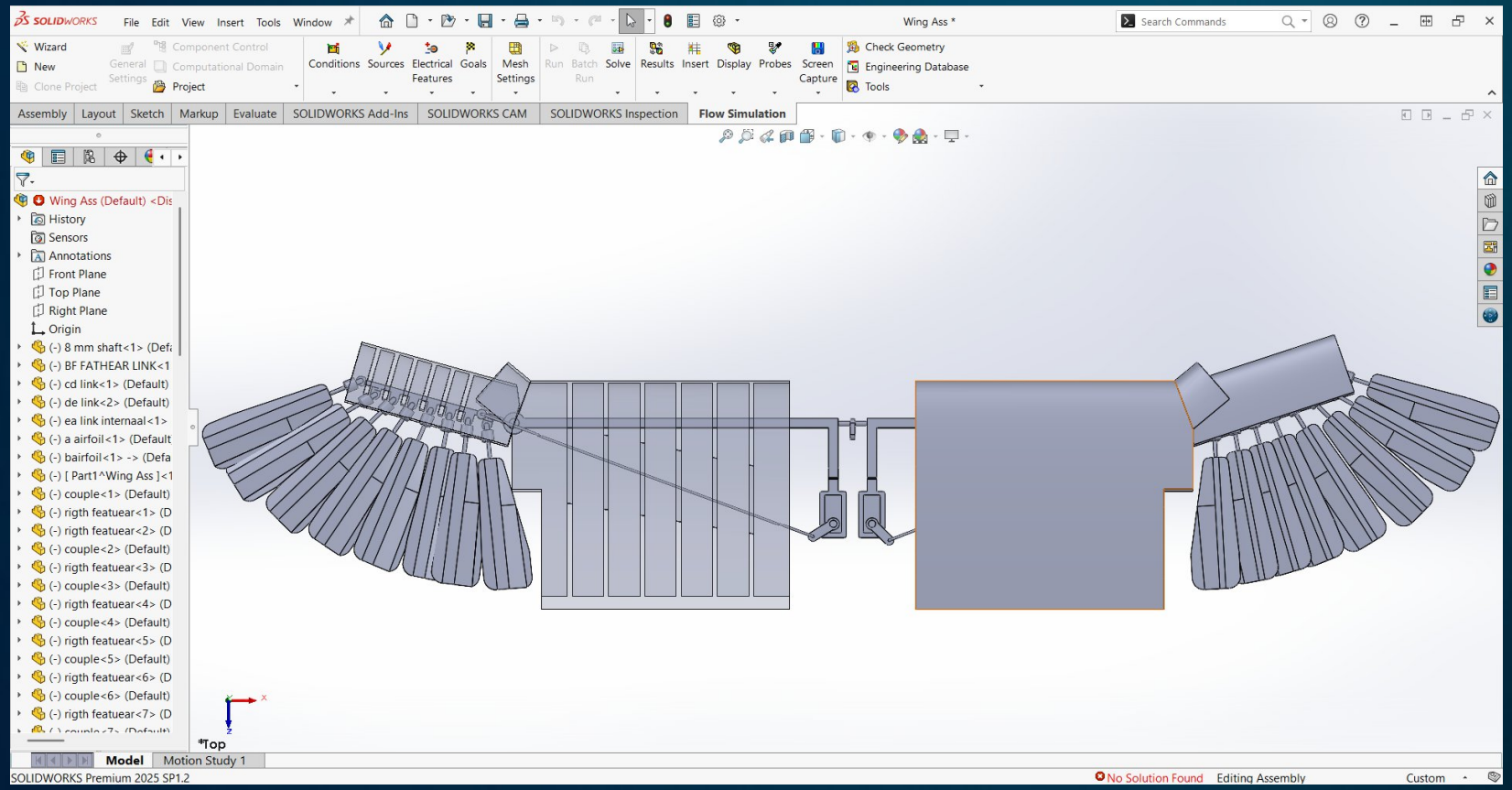
CAD Design



Wing mechanism

(First Prototype)
(Assembly)

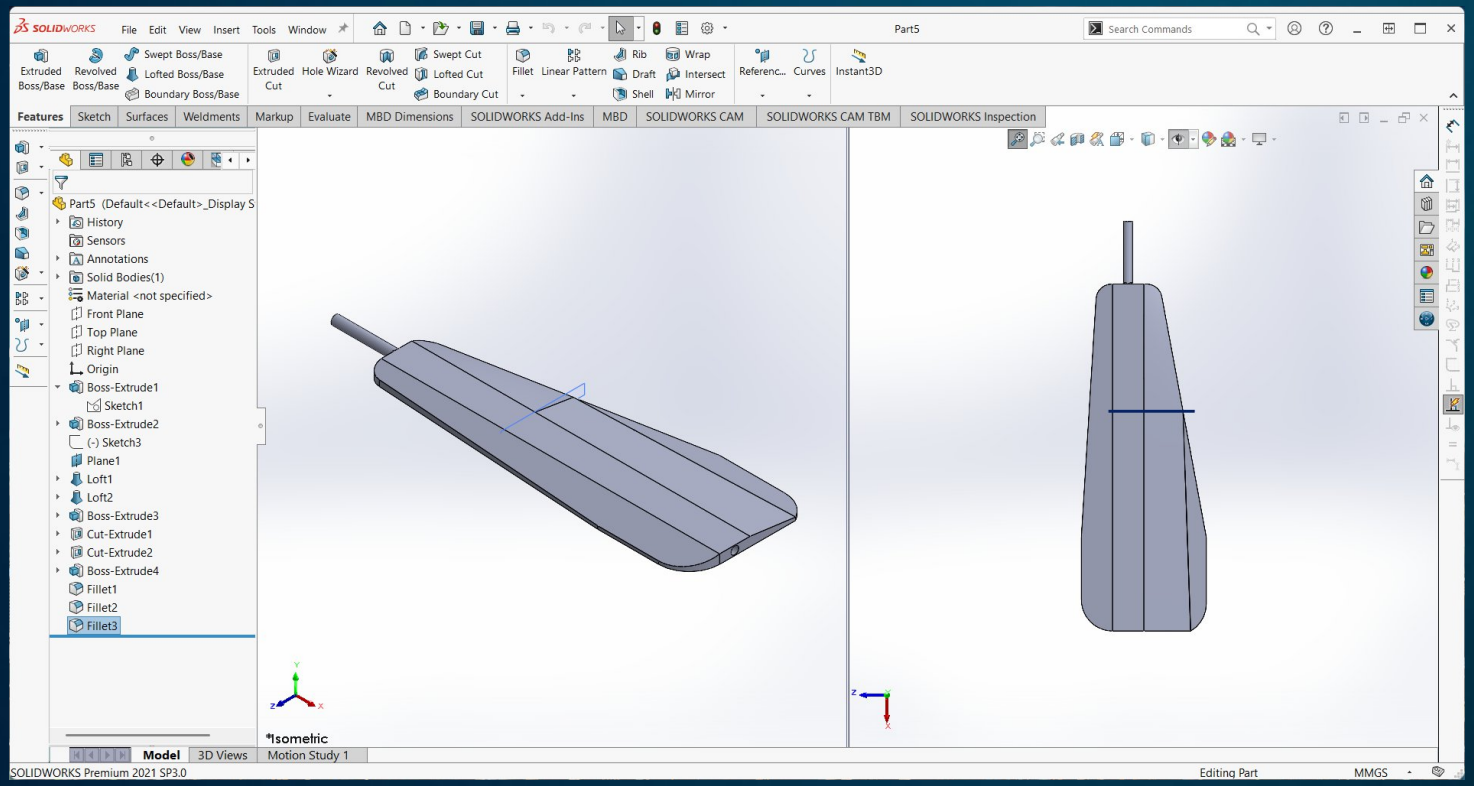
CAD Design



Wing mechanism

(New Design) (Assembly)

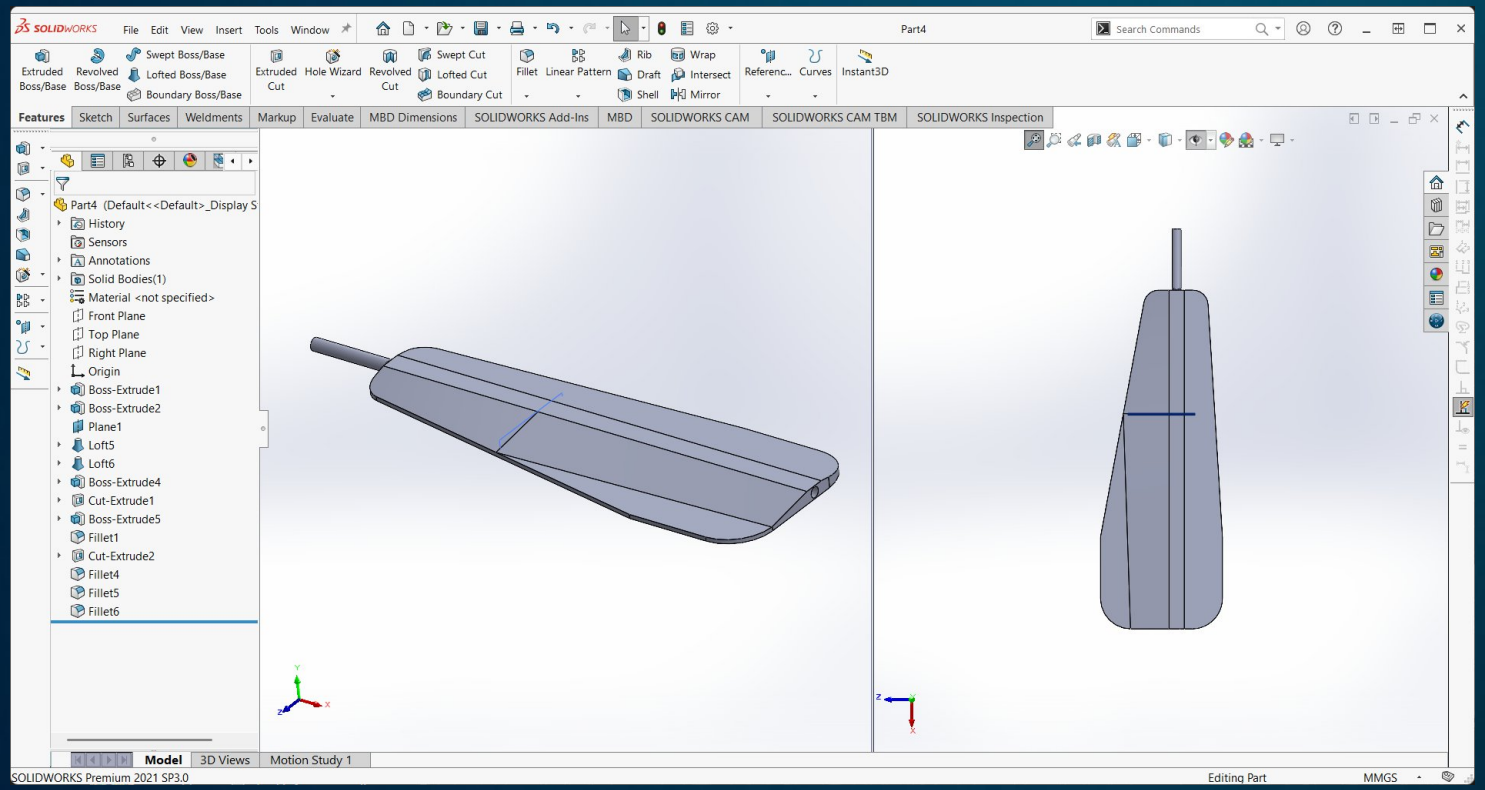
CAD Designs



Tail Feather (Left)

(Part Design)

CAD Designs



Tail Feather (Right)

(Part Design)

Tail Part Mechanism

The Tail Part mechanism consists of :

1. The Fan Mechanism (Morphing):
2. The Pitch & Yaw Mechanism (Elevator & Lateral Deflection)
3. The Twisting Mechanism (Rolling):

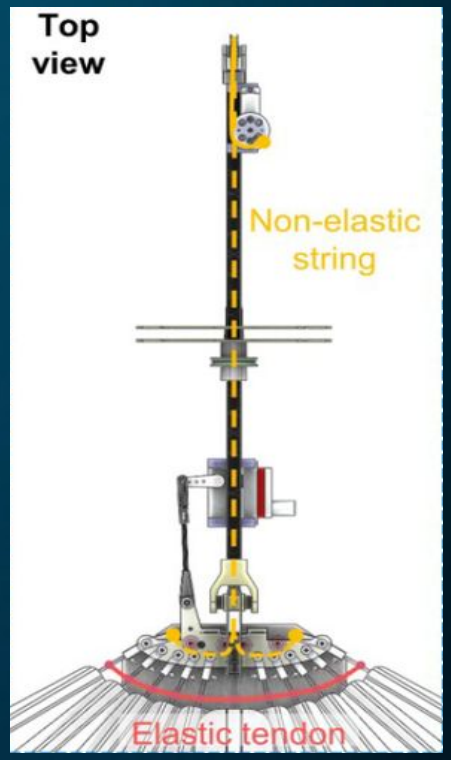
Mechanism (Links) that we Can Use for Achieving the flight in Air.

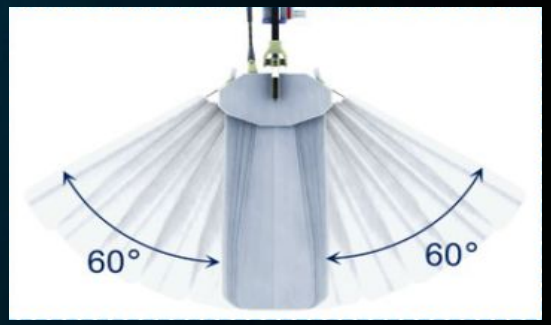
Mechanisms that Can be Used for Tail.

For The Tail Part We Need 3 Mechanisms To work Simultaneously to achieve the desire movement of tail :

1. The Fan Mechanism (Morphing):

This mechanism controls the spreading and closing of the feathers, much like a handheld fan. How it works: A servo pulls a non-elastic string (visible in detail b). This string is connected to the individual feathers. When the servo pulls, it overcomes the tension of an elastic tendon that runs across the base of the feathers. The Movement: It allows the tail to expand up to 60° on either side (detail c).Purpose: Spreading the tail increases lift and drag, helping the "bird" slow down or stabilize during flight.

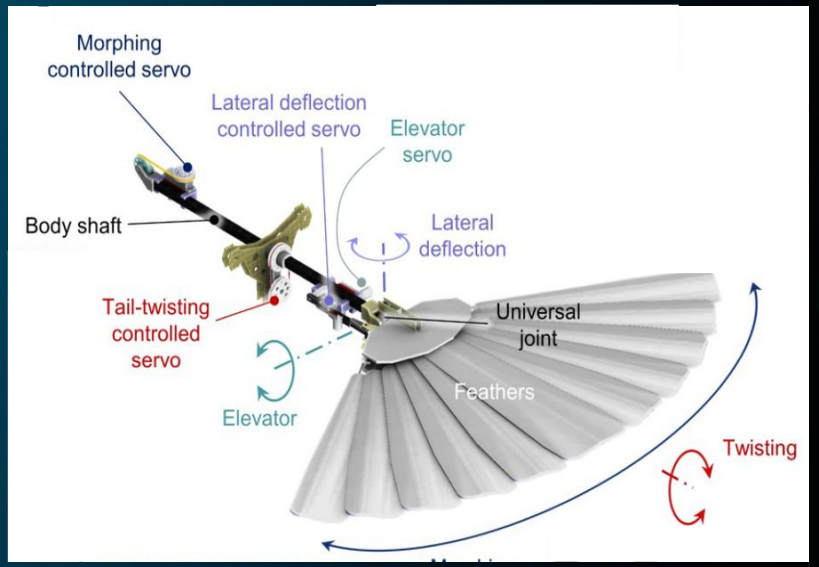




Links that we Can Use for Achieving the flight in Air.

Mechanisms that Can be Used for Tail.

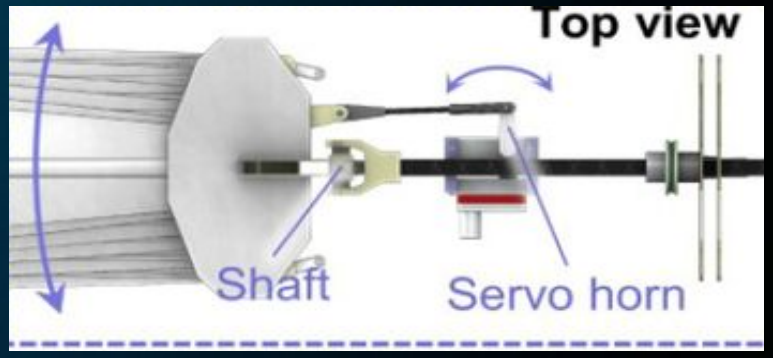
For The Tail Part We Need 3 Mechanisms To work Simultaneously to achieve the desire movement of tail :



2. The Pitch & Yaw Mechanism (Elevator & Lateral Deflection)

This system controls the **up-and-down** (Pitch) and **side-to-side** (Yaw) angling of the entire tail unit.

- **How it works:** It uses a **Universal Joint** (the gold-colored piece in detail This joint acts like a human wrist, allowing the tail to pivot in multiple directions.
  - **Elevator Servo:** Moves the tail up and down. This acts like the "flaps" on an airplane wing to control the climb or dive.
  - **Lateral Deflection Servo:** Swings the tail left or right

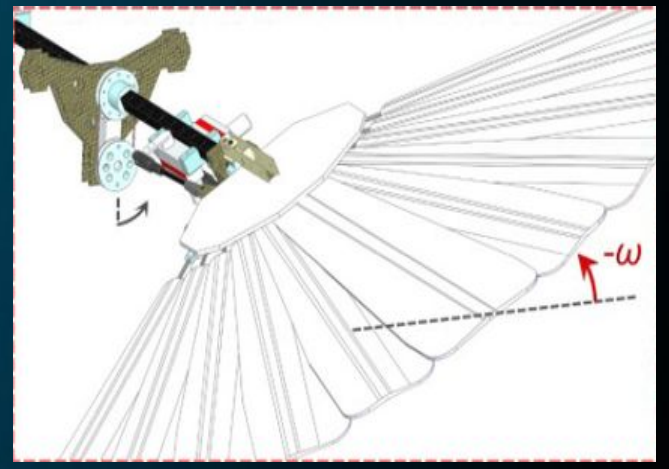


Purpose: This provides the primary steering, allowing the drone to turn or change altitude quickly.

Links that we Can Use for Achieving the flight in Air.

Mechanisms that Can be Used for Tail.

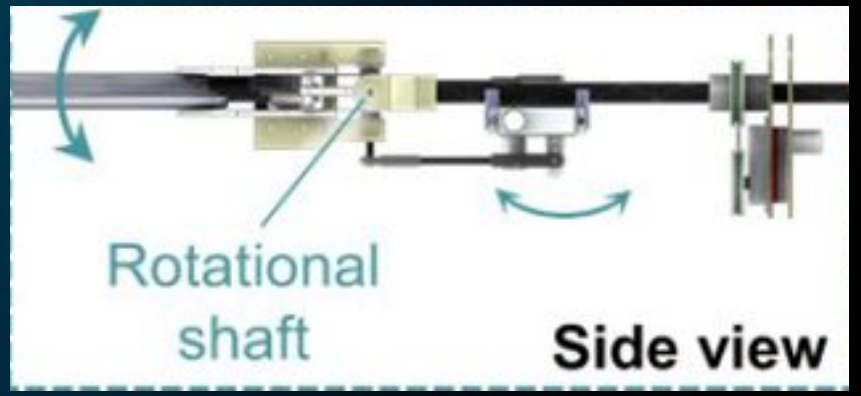
For The Tail Part We Need 3 Mechanisms To work Simultaneously to achieve the desire movement of tail :



3. The Twisting Mechanism (Rolling):

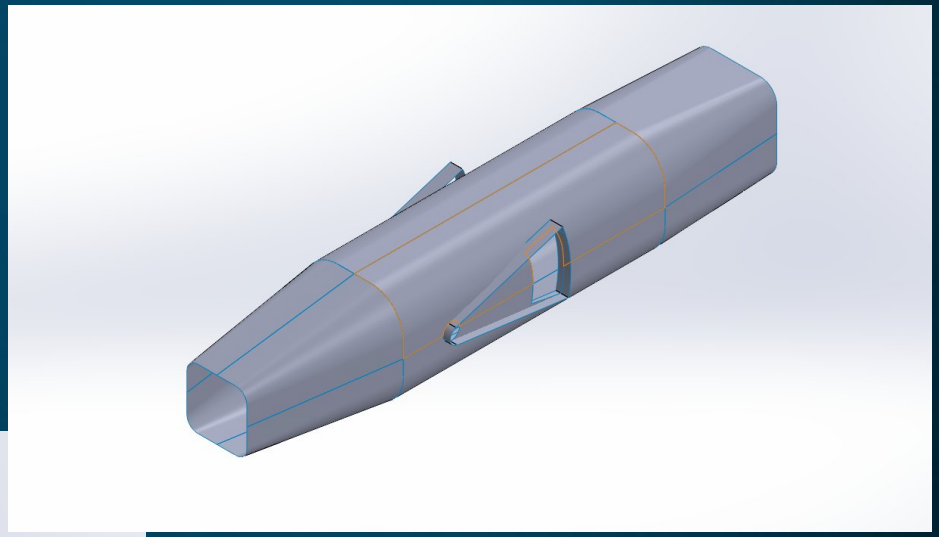
This is the most advanced part—it allows the tail to rotate or tilt along the axis of the body.

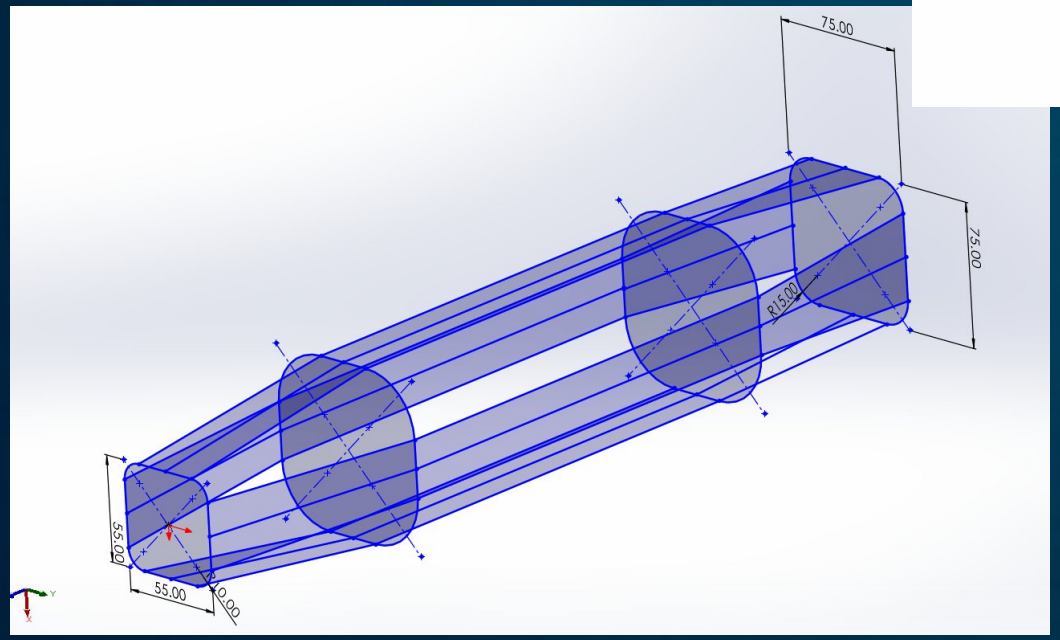
How it works: The Tail-twisting controlled servo is connected to a rotational shaft .Instead of just moving the tail up or left, it physically rotates the base of the tail assembly clockwise or counter-clockwise.



Purpose: By twisting the tail, the bird can perform "rolls" or stabilize itself against crosswinds. It mimics how real birds twist their tail feathers to maintain balance during turbulent flight.

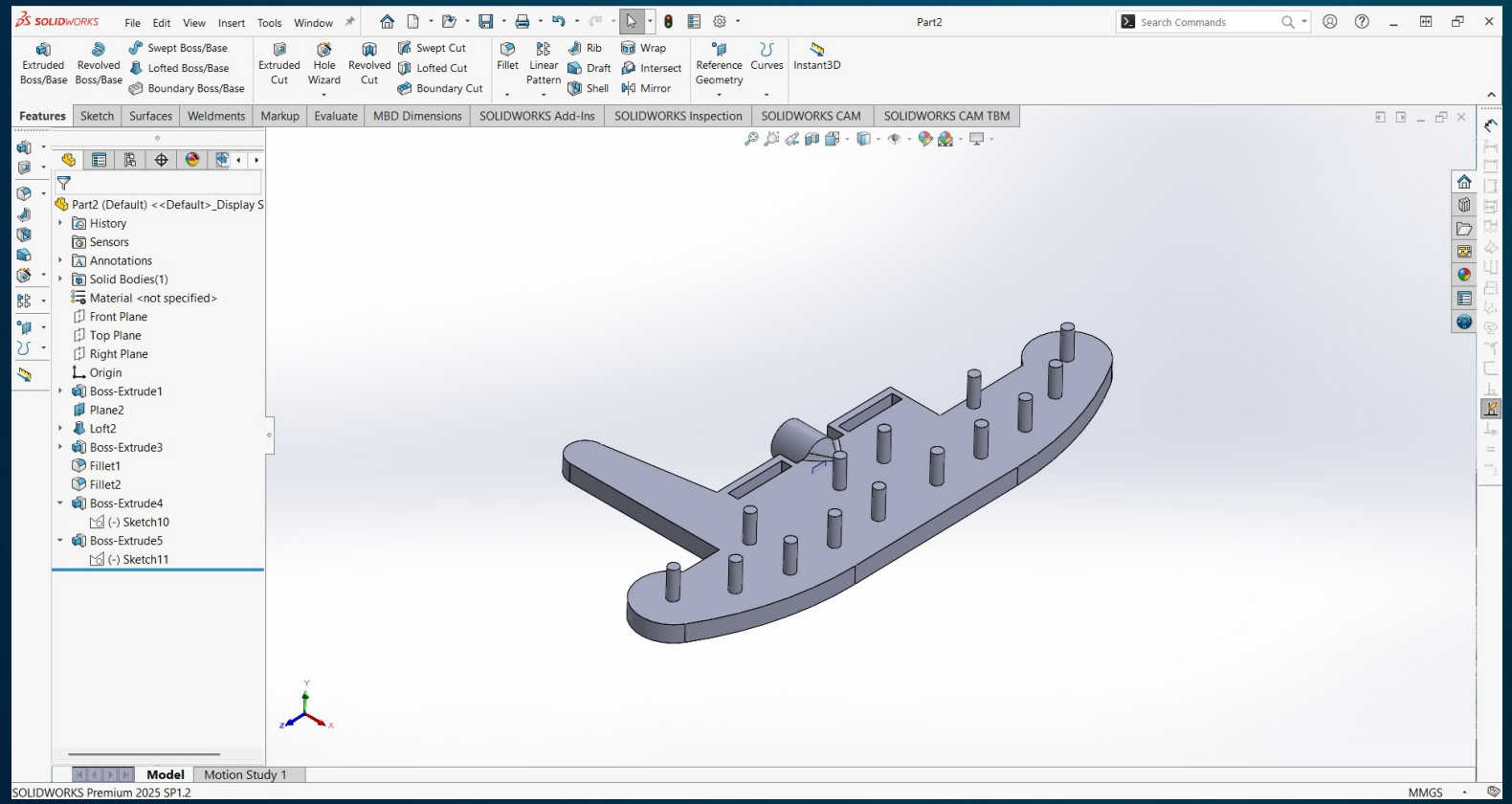
CAD Designs





Body
(Surface modeling)

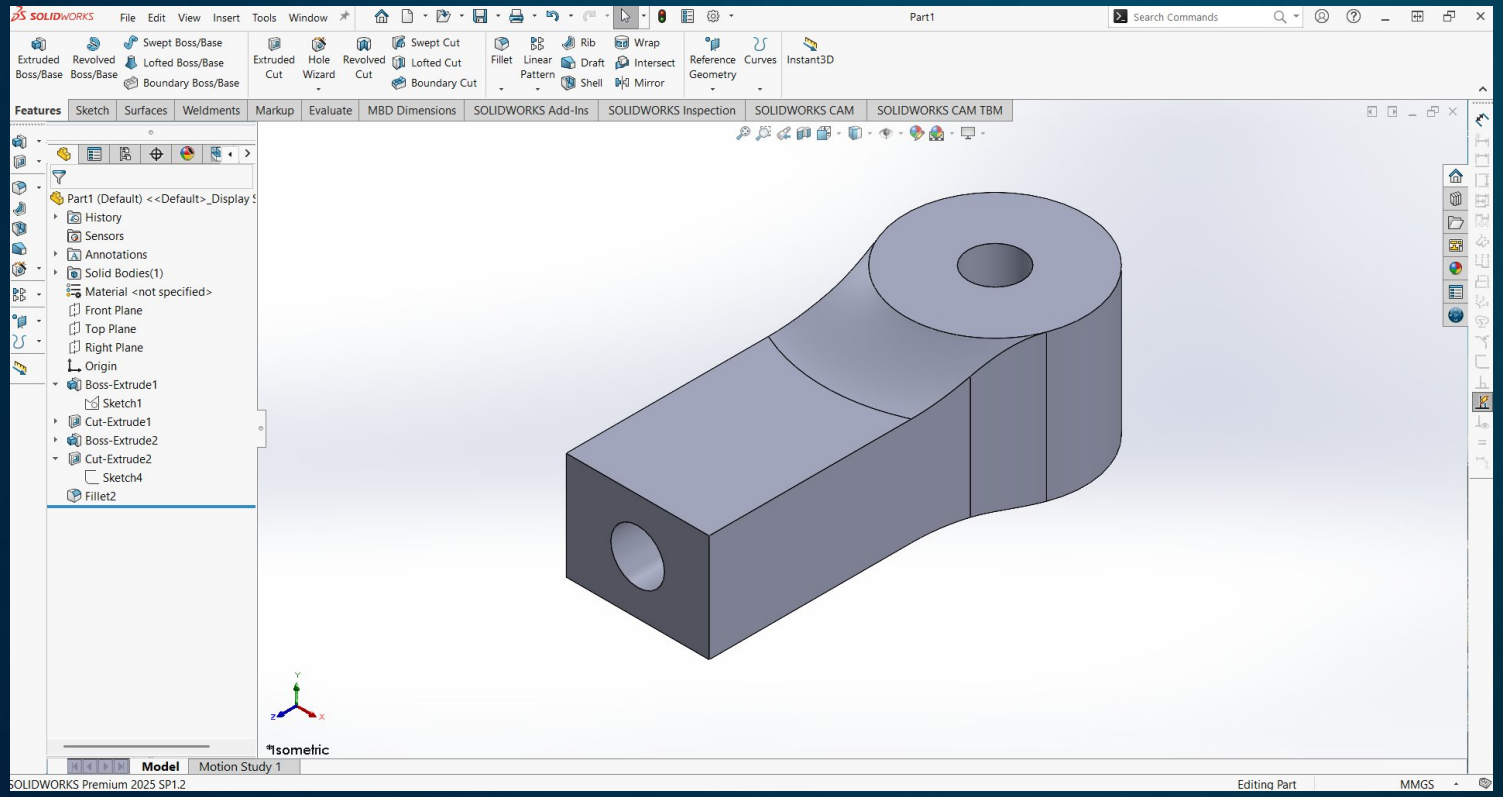
CAD Designs



Part 1

(Part Design)

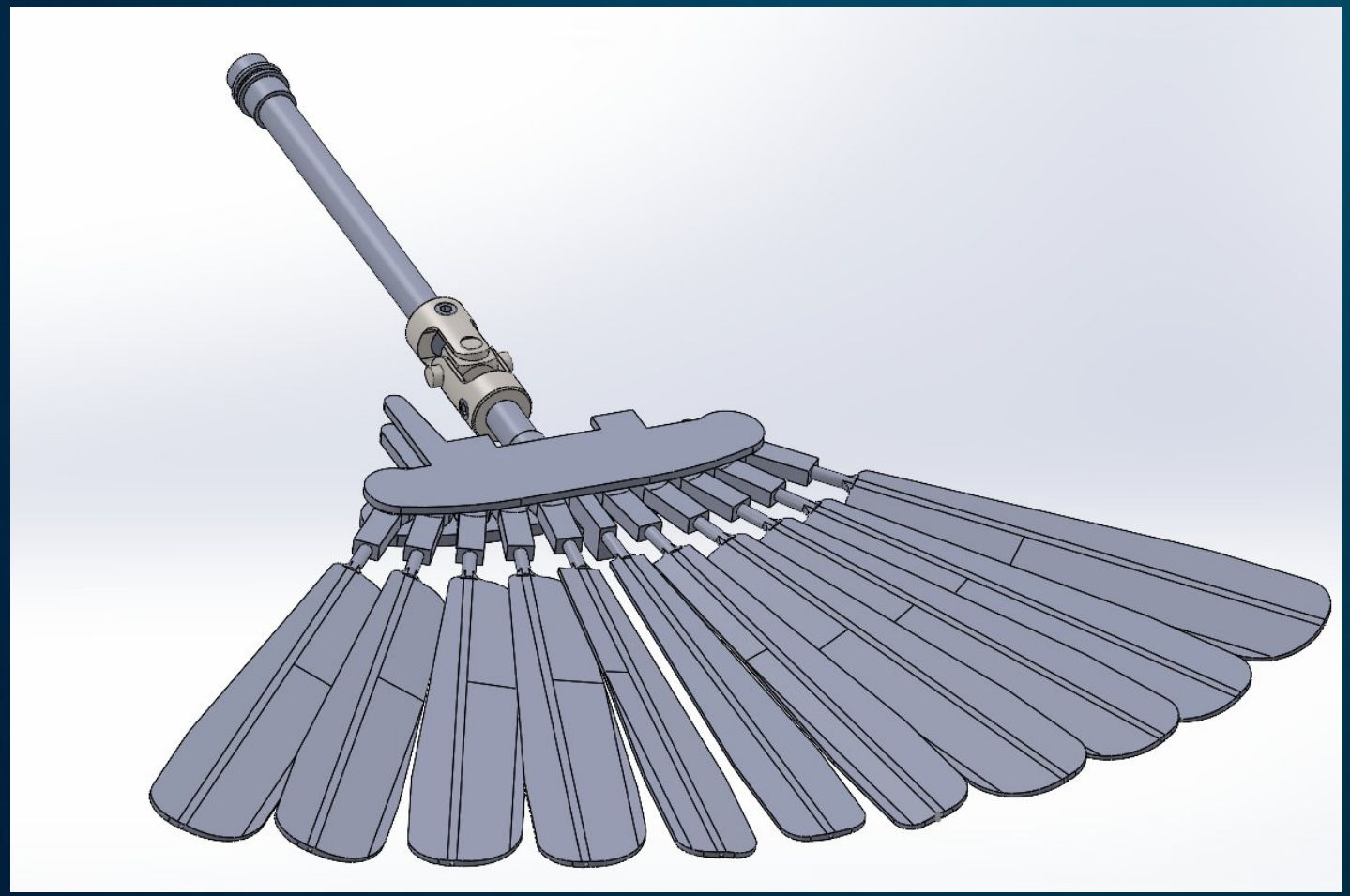
CAD Designs



Feather Holder

(Part Design)

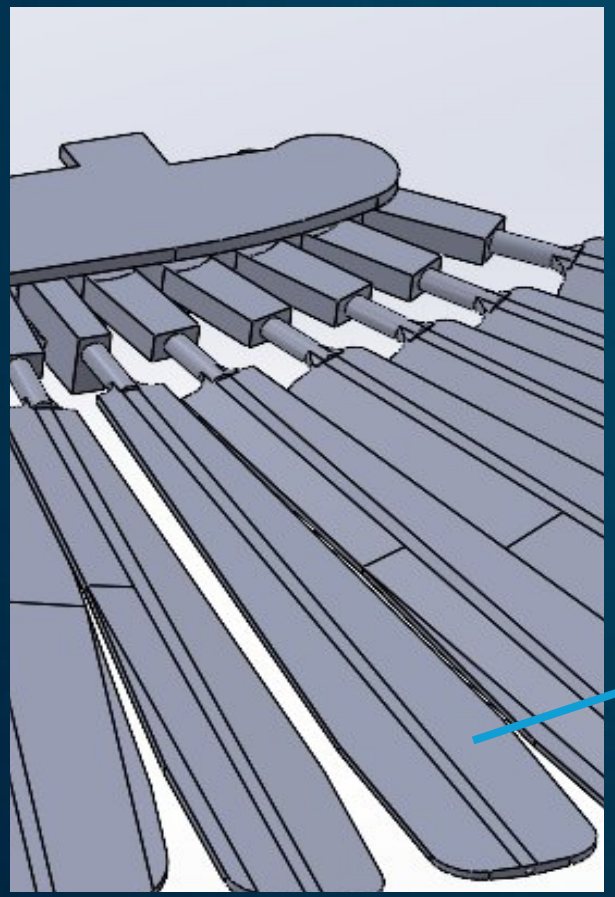
CAD Designs



Tail Part Mechanism

(Assembly)

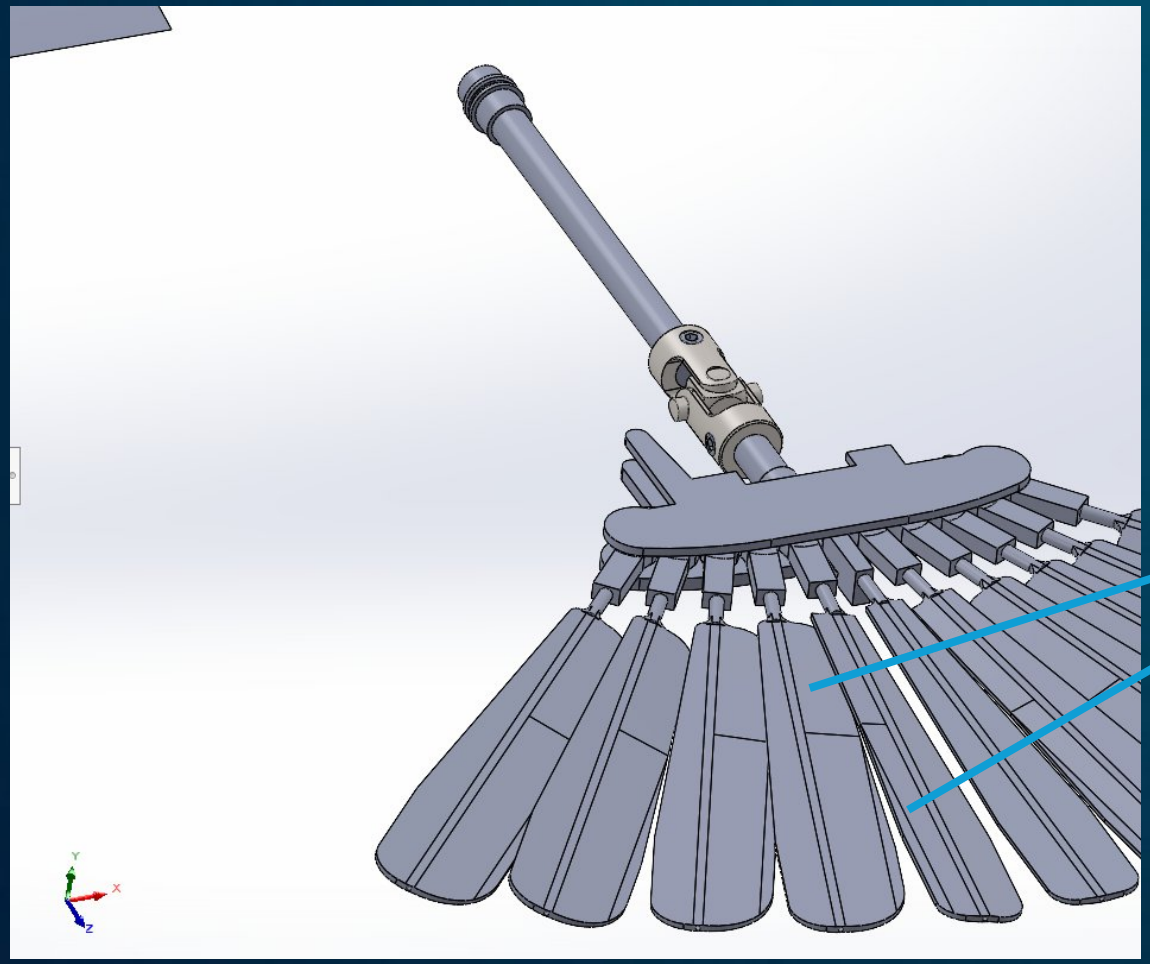
CAD Designs



Tail Feather (Center)

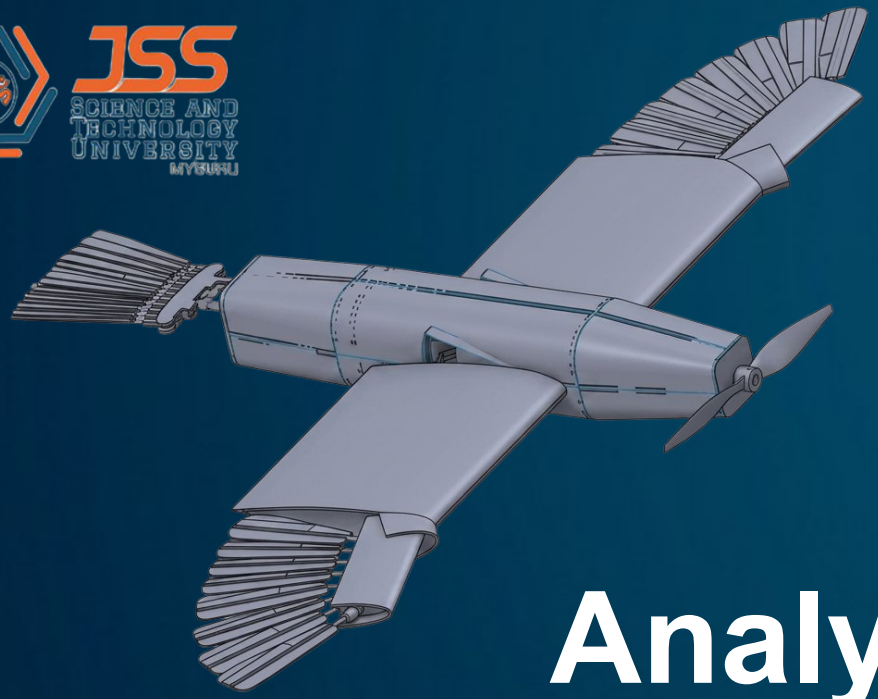
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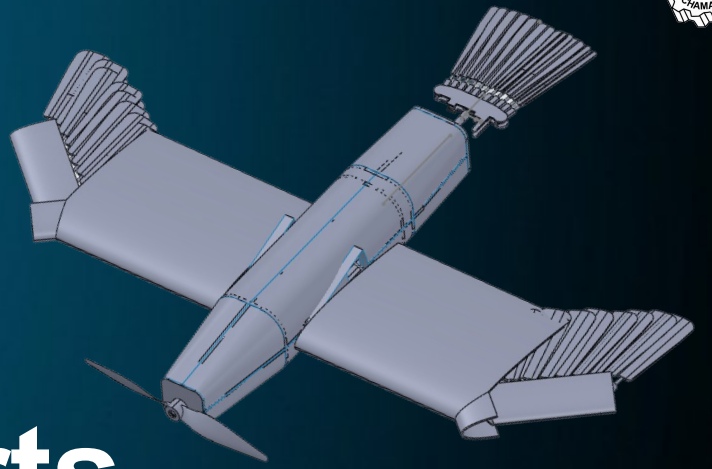
CAD Designs



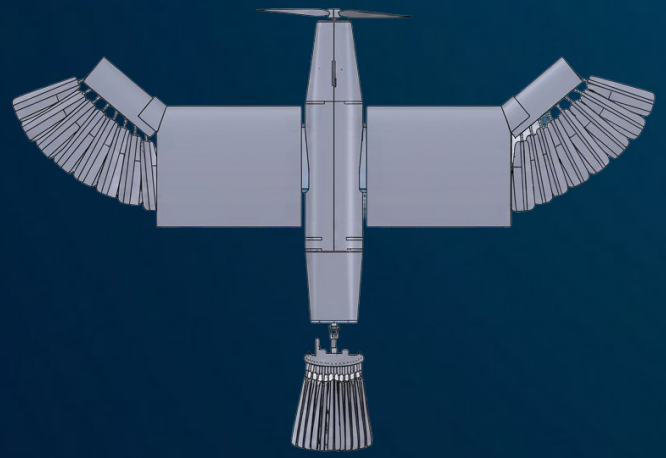
Tail Feather (Left)

(Part Design)





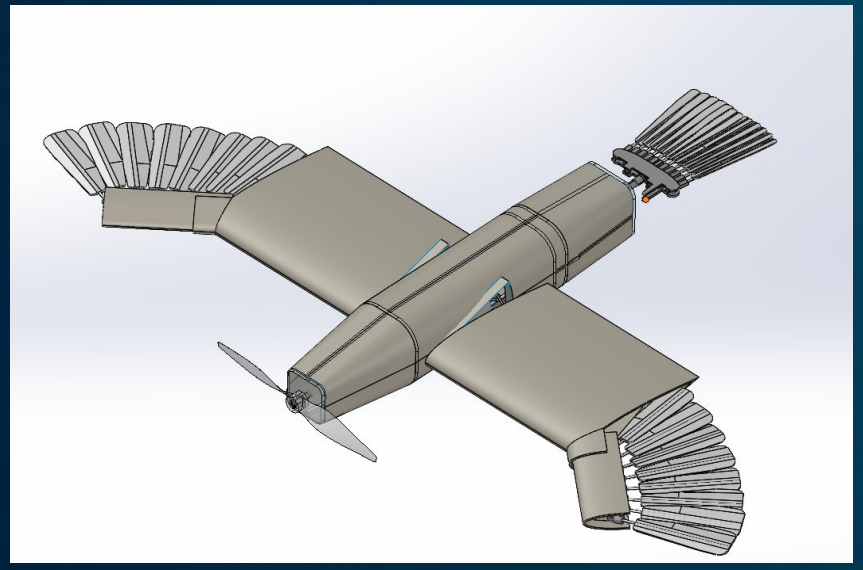
Analysis Reports

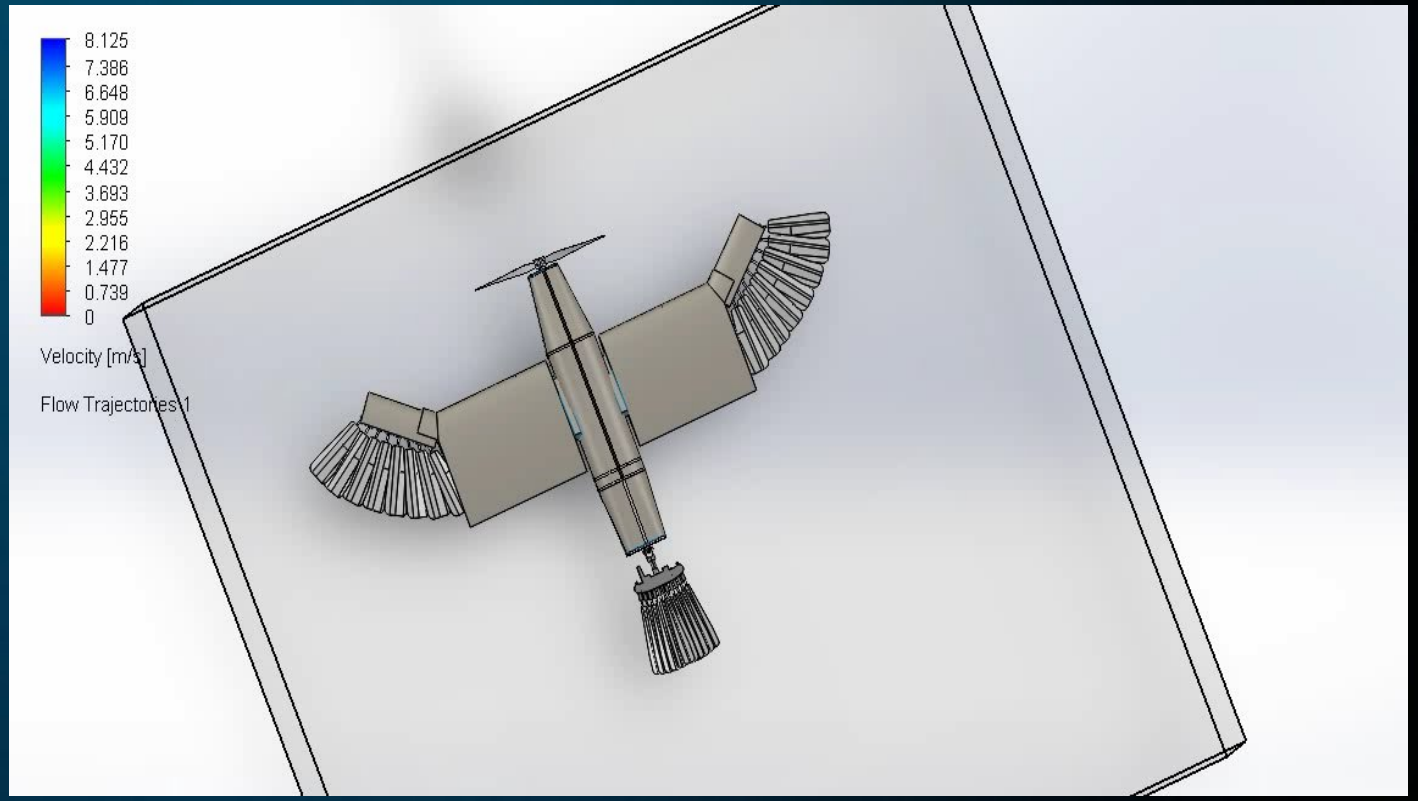




Analysis Reports

Condition 1: normal flight conditions





Analysis Reports

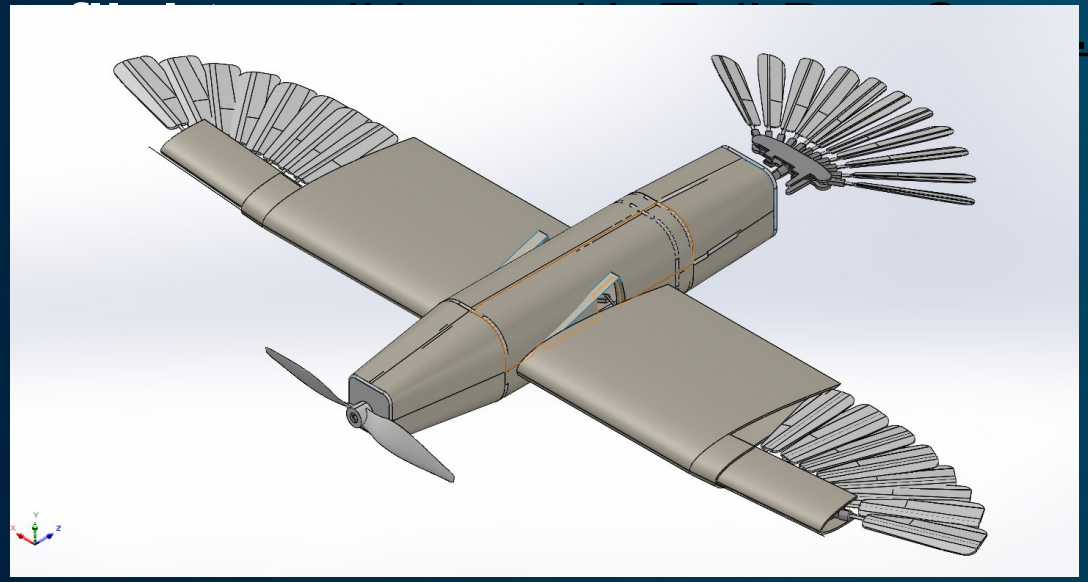
t1.SLDASM [T1 [Normal flight conditions]]

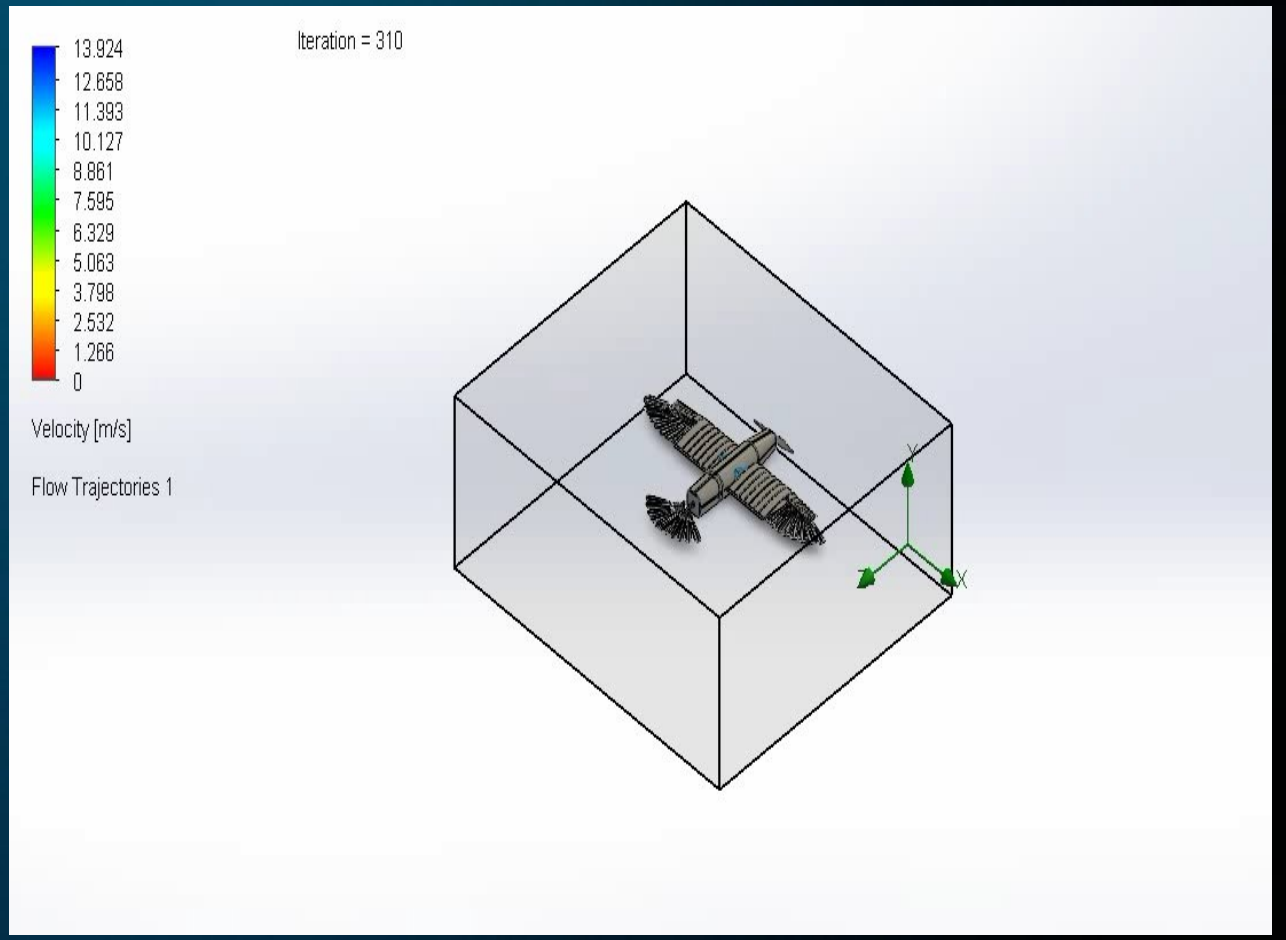
Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
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GG Average Total Pressure 2	[Pa]	101343.8702	101343.8704	101343.8688	101343.8715	100	Yes	0.000218132	0.010986493
GG Maximum Total Pressure 3	[Pa]	101365.0536	101364.958	101364.8276	101365.0983	100	Yes	0.014311043	31.9450551
GG Minimum Velocity 4	[m/s]	0	0	0	0	100	Yes	0	0
GG Average Velocity 5	[m/s]	6.921022499	6.921032828	6.920802696	6.921224405	100	Yes	3.28992E-05	0.00205691
GG Maximum Velocity 6	[m/s]	8.043165195	8.045462245	8.04216609	8.049710557	100	Yes	0.000418822	0.587696033
GG Minimum Velocity (X) 7	[m/s]	-2.22718708	-2.200461739	-2.22973101	-2.172105408	100	Yes	0.003915343	0.145620581
GG Average Velocity (X) 8	[m/s]	-0.00064194	-0.000642987	-0.000684114	-0.000615151	100	Yes	8.79189E-06	2.37658E-05
GG Maximum Velocity (X) 9	[m/s]	1.968217908	1.889661756	1.792840032	1.987614396	100	Yes	0.194774364	0.313587451
GG Minimum Velocity (Y) 10	[m/s]	-2.52489865	-2.529577477	-2.535413147	-2.524279987	100	Yes	0.00045257	0.015924533
GG Average Velocity (Y) 11	[m/s]	0.001186363	0.001235318	0.001122332	0.001319453	100	Yes	3.98943E-05	4.30363E-05
GG Maximum Velocity (Y) 12	[m/s]	2.471280537	2.438163199	2.392712706	2.483661017	100	Yes	0.003347012	0.015405095
GG Minimum Velocity (Z) 13	[m/s]	-6.34500757	-6.422063117	-6.528293204	-6.321889006	100	Yes	0.185201355	0.652071777
GG Average Velocity (Z) 14	[m/s]	6.917796854	6.917829495	6.917580025	6.918010566	100	Yes	3.12755E-05	0.002055678
GG Maximum Velocity (Z) 15	[m/s]	8.009918871	8.014311322	8.00894517	8.022903841	100	Yes	0.000895887	0.033915299
GG Force (Y) 16	[N]	0.013222579	0.013521146	0.011498545	0.015913094	100	Yes	0.00035637	0.003618832
GG Friction Force (Z) 17	[N]	0.033713151	0.033646564	0.033449563	0.033880725	100	Yes	3.5167E-05	0.000487413

Iterations [ ]: 295
Analysis interval: 39

Analysis Reports

Condition 1: normal flight conditions with Tail Part Open.





Analysis Reports

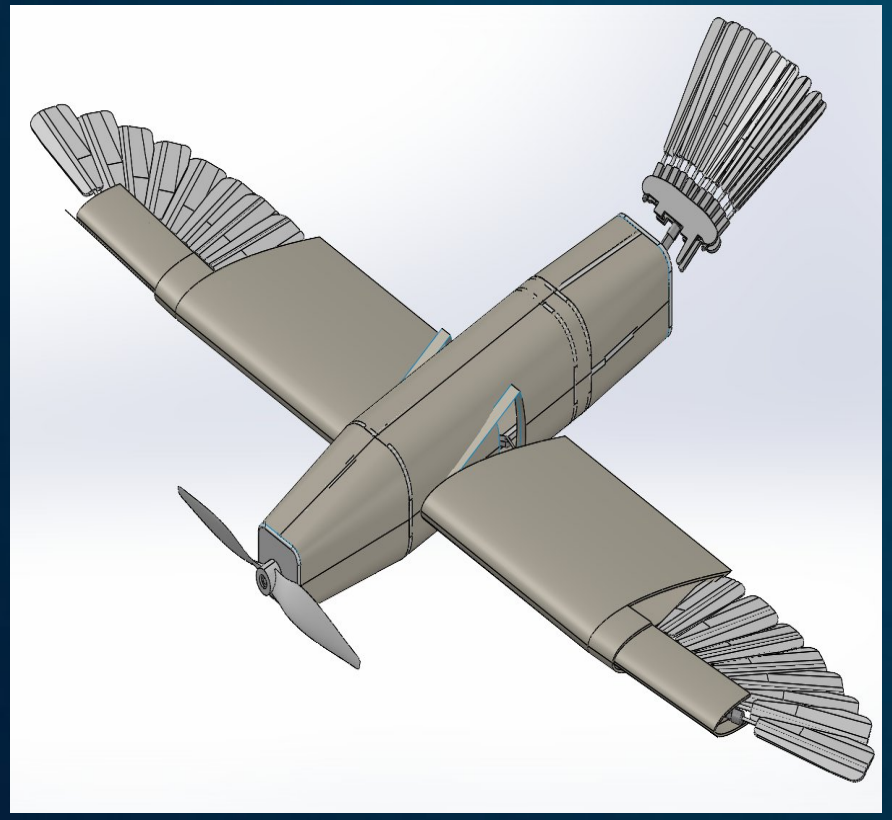
t2 xyz.SLDASM [T2 [Normal flight conditions with Tail Part Open.]]

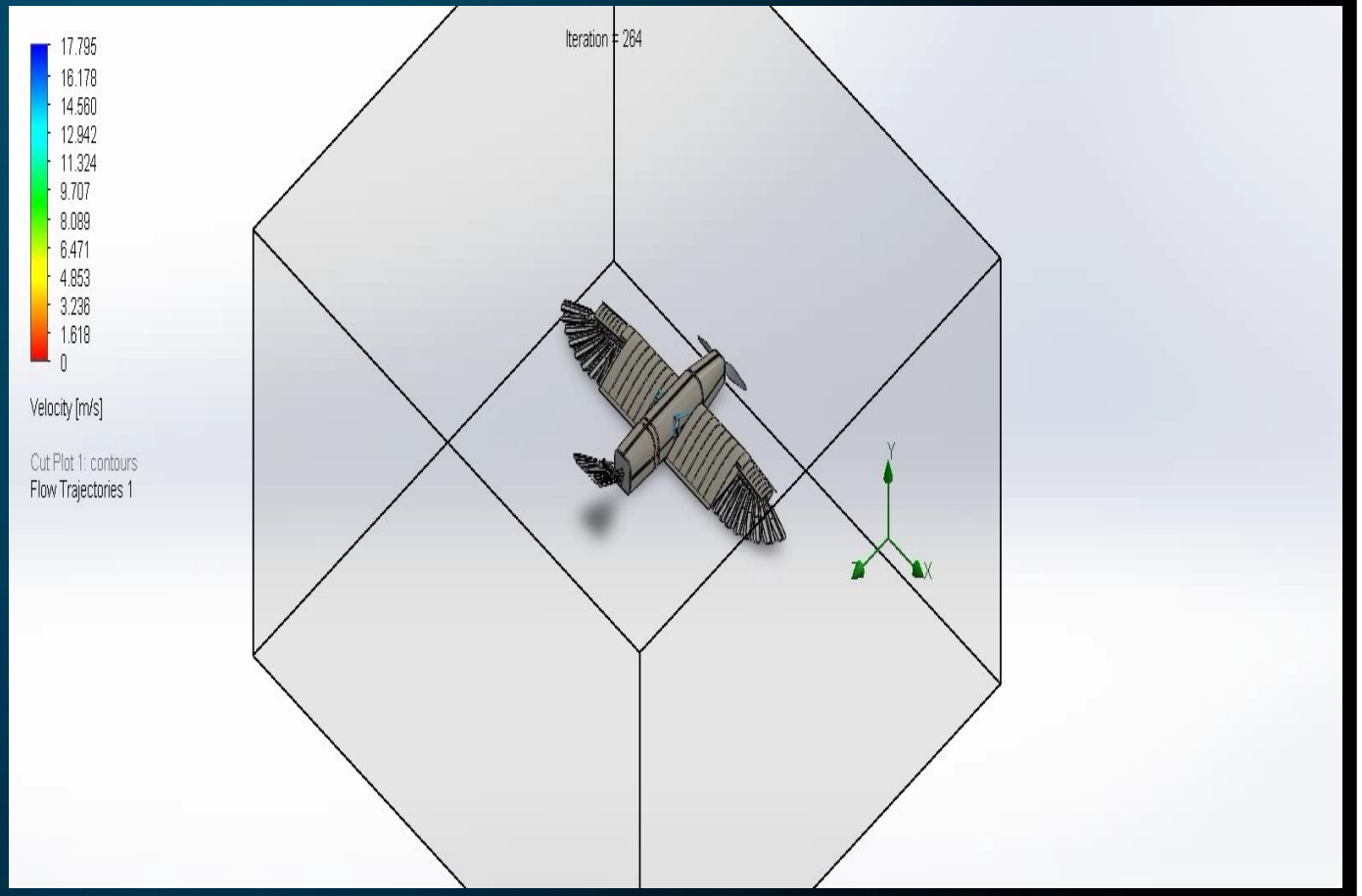
Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
GG Minimum Total Pressure 1	[Pa]	100991.8078	100991.0193	100989.3024	100992.3635	100	Yes	3.061172093	94.20389942
GG Average Total Pressure 2	[Pa]	101398.3112	101398.3179	101398.3058	101398.3263	100	Yes	0.001549848	0.032357321
GG Maximum Total Pressure 3	[Pa]	101460.6385	101460.5179	101460.3709	101460.6582	100	Yes	0.051814521	125.311799
GG Minimum Velocity 4	[m/s]	0	0	0	0	100	Yes	0	0
GG Average Velocity 5	[m/s]	11.86534555	11.86590551	11.86490247	11.86668162	100	Yes	0.000126238	0.003586778
GG Maximum Velocity 6	[m/s]	16.11014914	16.8884322	16.11014914	17.69251094	24.9	Yes	1.582361803	0.394073381
GG Minimum Velocity (X) 7	[m/s]	-4.6229257	-4.56242163	-5.665434024	-4.033228474	100	Yes	0.158496833	0.175304715
GG Average Velocity (X) 8	[m/s]	0.000109474	0.000236139	4.01905E-05	0.000358542	100	Yes	2.33842E-05	4.09367E-05
GG Maximum Velocity (X) 9	[m/s]	4.28511868	4.286903082	4.197577691	4.460589131	100	Yes	0.024415837	0.136417947
GG Minimum Velocity (Y) 10	[m/s]	-4.26802455	-4.241644132	-4.268024552	-4.217123181	100	Yes	0.006733881	0.066858188
GG Average Velocity (Y) 11	[m/s]	0.000695045	0.000970026	0.000695045	0.00127543	71.8	Yes	0.00014651	0.00010523
GG Maximum Velocity (Y) 12	[m/s]	4.256364189	4.224264212	4.190440205	4.25804148	100	Yes	0.003744361	0.020020056
GG Minimum Velocity (Z) 13	[m/s]	-16.1087895	-16.8869781	-17.69102645	-16.10878948	30.9	Yes	1.582236977	0.489822116
GG Average Velocity (Z) 14	[m/s]	11.85869934	11.8592845	11.85829842	11.85999825	100	Yes	0.000135678	0.003618
GG Maximum Velocity (Z) 15	[m/s]	13.8847801	13.84606368	13.78996649	13.89866931	100	Yes	0.009339928	0.028573612
GG Force (Y) 16	[N]	0.034198297	0.026064605	0.017953286	0.034198297	100	Yes	0.003651215	0.010049979
GG Friction Force (Z) 17	[N]	0.072460256	0.073272882	0.07243846	0.074148247	100	Yes	0.000154108	0.001616894

Iterations [ ]: 310
Analysis interval: 39

Analysis Reports

Condition 1: LIFT Condition





Analysis Reports

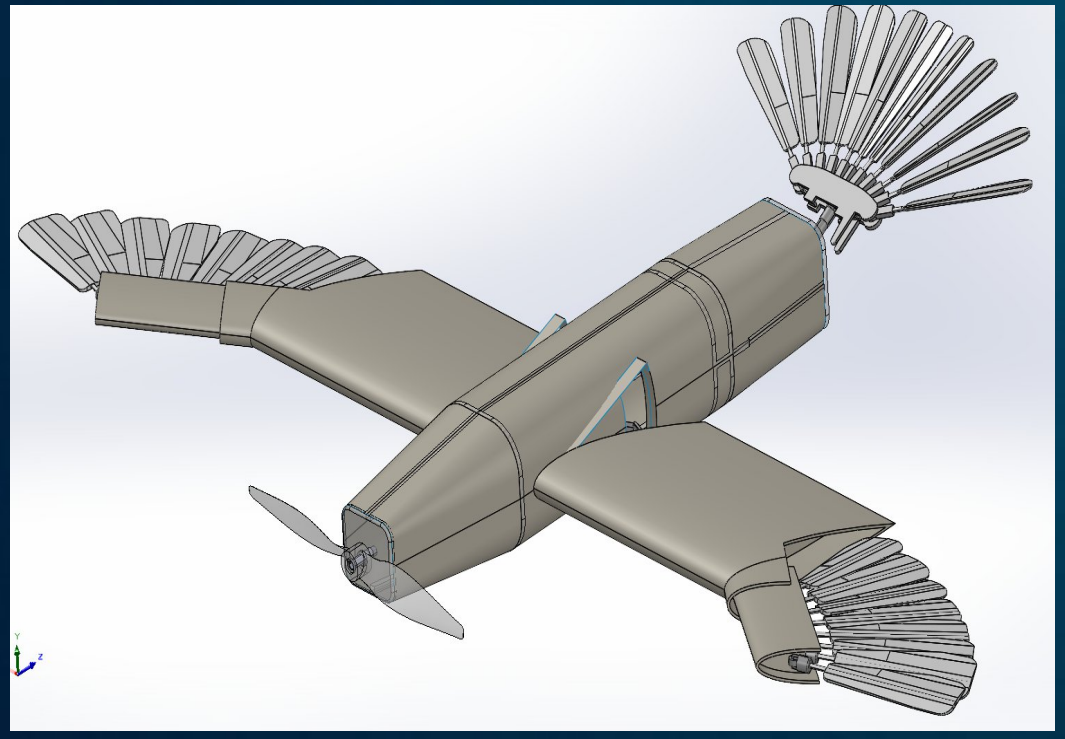
t3 xyz.SLDASM [t3 [LIFT Condition]]

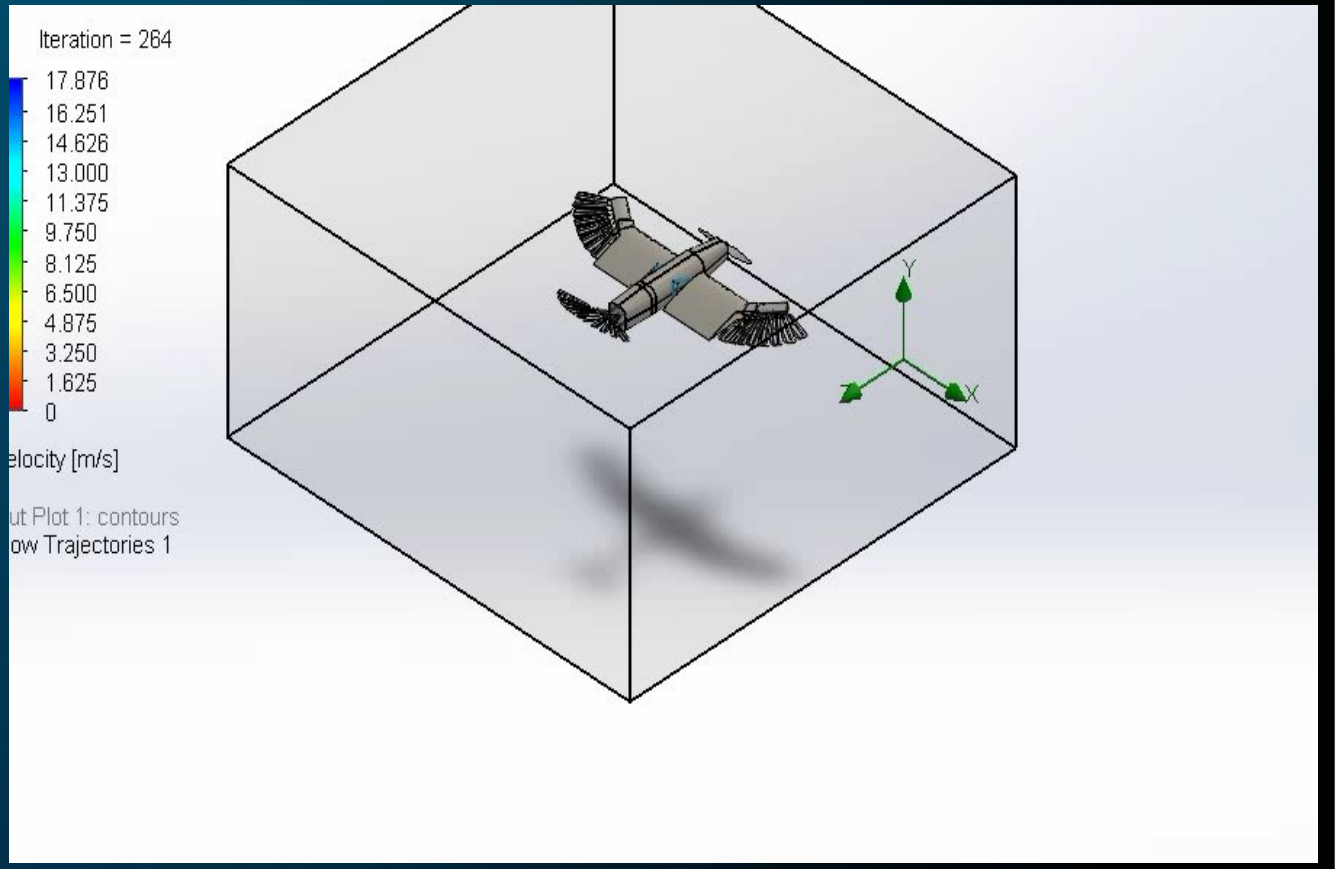
Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
GG Minimum Total Pressure 1	[Pa]	101227.8563	101228.3478	101227.4019	101229.7765	100	Yes	0.954573044	95.85948305
GG Average Total Pressure 2	[Pa]	101463.2291	101463.2299	101463.2243	101463.2338	100	Yes	0.002346122	0.034872391
GG Maximum Total Pressure 3	[Pa]	101620.8163	101622.5444	101620.0529	101626.2003	100	Yes	5.822422621	141.7470001
GG Minimum Velocity 4	[m/s]	0	0	0	0	100	Yes	0	0
GG Average Velocity 5	[m/s]	15.90755634	15.90763756	15.90713337	15.90797449	100	Yes	0.000268421	0.002480727
GG Maximum Velocity 6	[m/s]	18.04968777	17.99528137	17.92768799	18.07910437	100	Yes	0.071131376	0.524852203
GG Minimum Velocity (X) 7	[m/s]	-5.39599648	-5.514341753	-5.701977144	-5.341200704	100	Yes	0.152132798	0.195445382
GG Average Velocity (X) 8	[m/s]	-6.088E-05	-2.69189E-05	-8.57257E-05	3.93325E-05	51.8	Yes	3.09978E-05	1.6068E-05
GG Maximum Velocity (X) 9	[m/s]	5.561503369	5.554996088	5.468285759	5.62183689	100	Yes	0.063929731	0.224065643
GG Minimum Velocity (Y) 10	[m/s]	-8.60340066	-8.837037382	-9.310456416	-8.588100834	18.3	Yes	0.248540502	0.045669777
GG Average Velocity (Y) 11	[m/s]	-0.06001403	-0.060032732	-0.06026695	-0.059693084	100	Yes	0.000386132	0.001789494
GG Maximum Velocity (Y) 12	[m/s]	3.816996717	3.710747879	3.368170001	4.291196813	36.7	Yes	0.108806585	0.04002855
GG Minimum Velocity (Z) 13	[m/s]	-5.50273253	-5.699108807	-5.906899941	-5.494970376	100	Yes	0.039242333	0.894100563
GG Average Velocity (Z) 14	[m/s]	15.90261659	15.90259048	15.90218976	15.90282571	100	Yes	0.000206766	0.002523429
GG Maximum Velocity (Z) 15	[m/s]	18.04751305	17.98930573	17.91881415	18.0766875	26.8	Yes	0.068737558	0.018457754
GG Force (Y) 16	[N]	3.153456679	3.162389184	3.150795867	3.171494002	100	Yes	0.015616232	0.15520379
GG Friction Force (Z) 17	[N]	0.0895307	0.089905985	0.087579437	0.091834709	100	Yes	0.000337462	0.002755378

Iterations [ ]: 264
Analysis interval: 33

Analysis Reports

Condition 1: LIFT Condition with Tail Open





Analysis Reports

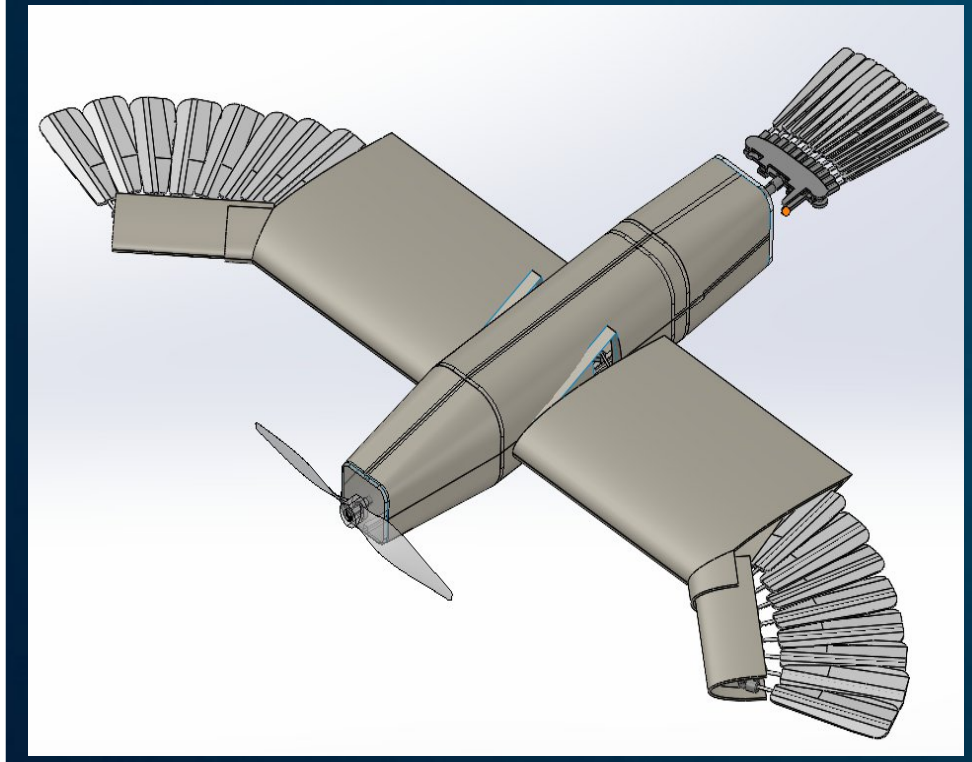
nt4.SLDASM [T4 [LIFT Condition with Tail Open]]

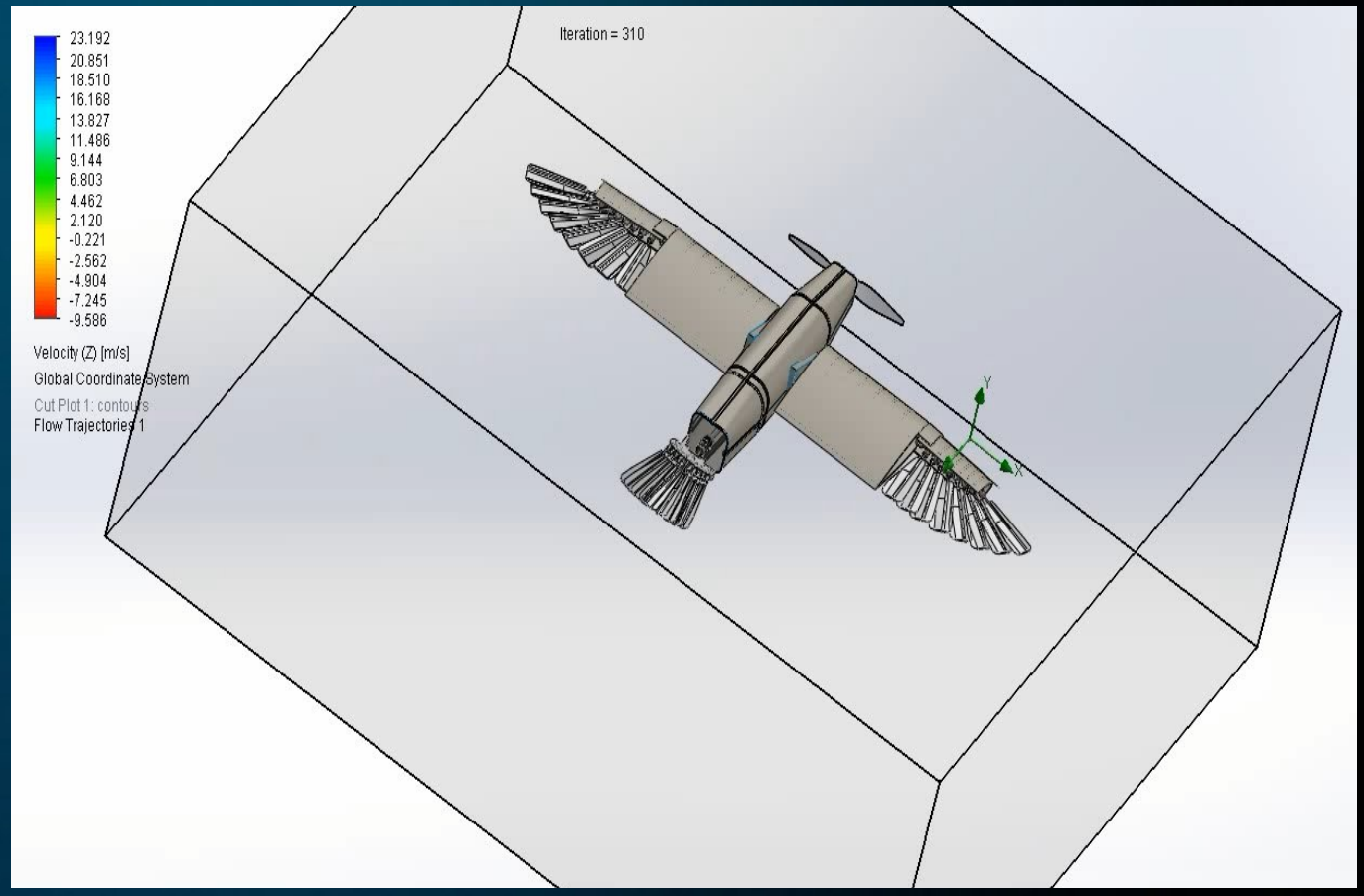
Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
GG Minimum Total Pressure 1	[Pa]	101219.9813	101221.8869	101218.2353	101224.32	100	Yes	0.606215731	72.46030834
GG Average Total Pressure 2	[Pa]	101463.1815	101463.1909	101463.1815	101463.201	100	Yes	0.007675837	0.036722858
GG Maximum Total Pressure 3	[Pa]	101636.8584	101630.937	101613.8797	101641.0617	100	Yes	6.641240104	112.6154827
GG Minimum Velocity 4	[m/s]	0	0	0	0	100	Yes	0	0
GG Average Velocity 5	[m/s]	15.90272414	15.90305806	15.90249731	15.90358444	100	Yes	0.000654504	0.002633795
GG Maximum Velocity 6	[m/s]	18.13266282	18.16777068	18.01125269	18.29311391	100	Yes	0.048104619	0.079731733
GG Minimum Velocity (X) 7	[m/s]	-5.56387081	-5.211015522	-5.710784143	-4.613771079	100	Yes	0.137935768	0.226419015
GG Average Velocity (X) 8	[m/s]	-0.00131689	-0.001184156	-0.001339408	-0.000980525	100	Yes	4.64531E-05	8.73023E-05
GG Maximum Velocity (X) 9	[m/s]	4.902950298	4.762416056	4.514614001	5.45522464	100	Yes	0.05759685	0.097672357
GG Minimum Velocity (Y) 10	[m/s]	-9.74636948	-9.447149927	-10.26707986	-8.72864862	26.9	Yes	0.274830779	0.074171276
GG Average Velocity (Y) 11	[m/s]	-0.06341542	-0.063804994	-0.064504879	-0.063259322	100	Yes	0.000584143	0.001861516
GG Maximum Velocity (Y) 12	[m/s]	3.914517211	3.77202361	3.184314433	4.565039538	57.7	Yes	0.161773282	0.093480443
GG Minimum Velocity (Z) 13	[m/s]	-2.00248234	-2.33344986	-2.752787152	-1.876561137	100	Yes	0.11642899	0.366470047
GG Average Velocity (Z) 14	[m/s]	15.89728509	15.89767616	15.89710372	15.89822897	100	Yes	0.000601166	0.002688879
GG Maximum Velocity (Z) 15	[m/s]	18.10917546	18.1458473	17.98470495	18.2635022	100	Yes	0.04708076	0.076060536
GG Force (Y) 16	[N]	2.845841586	2.86758899	2.845059122	2.893252358	100	Yes	0.028116179	0.196040907
GG Friction Force (Z) 17	[N]	0.103975404	0.104409241	0.1015318	0.106961508	100	Yes	0.000334517	0.002527143

Iterations [ ]: 264
Analysis interval: 33

Analysis Reports

Condition 1: CRUISING Condition





Analysis Reports

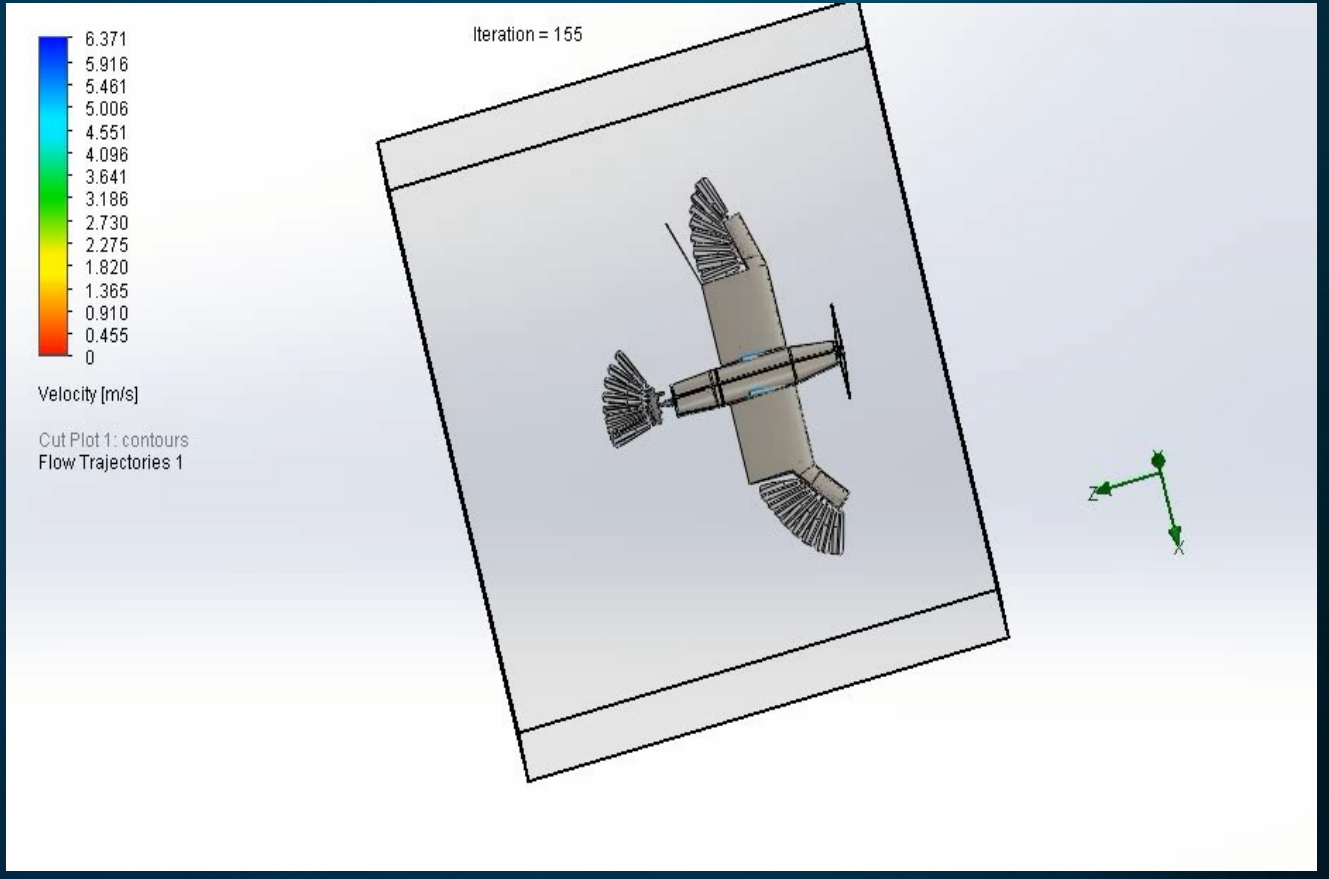
nt5.SLDASM [t5 [CRUISING Condition]]

Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
GG Minimum Total Pressure 1	[Pa]	100964.0861	100960.935	100955.5314	100965.3104	100	Yes	0.630108929	236.9094803
GG Average Total Pressure 2	[Pa]	101545.1942	101545.1924	101545.1713	101545.2081	100	Yes	0.004015279	0.087110488
GG Maximum Total Pressure 3	[Pa]	101718.6469	101718.3331	101717.8037	101718.7809	100	Yes	0.140788998	259.9310024
GG Minimum Velocity 4	[m/s]	0	0	0	0	100	Yes	0	0
GG Average Velocity 5	[m/s]	19.77895052	19.77876478	19.77761589	19.77962712	100	Yes	0.000264183	0.005713632
GG Maximum Velocity 6	[m/s]	29.43942758	31.04810746	29.43942758	32.74189194	29.8	Yes	3.302464362	0.98652408
GG Minimum Velocity (X) 7	[m/s]	-7.21921504	-7.690934979	-8.36543993	-7.046283902	100	Yes	0.188215735	0.40229458
GG Average Velocity (X) 8	[m/s]	0.000487869	0.000504502	0.000311564	0.000642478	100	Yes	3.51221E-05	6.16384E-05
GG Maximum Velocity (X) 9	[m/s]	7.148264144	7.429225094	7.126123337	8.493189776	100	Yes	0.064996477	0.422867011
GG Minimum Velocity (Y) 10	[m/s]	-7.0915087	-7.126946733	-7.412239968	-7.074731981	48.2	Yes	0.037669349	0.018175096
GG Average Velocity (Y) 11	[m/s]	0.002057169	0.002321032	0.002051281	0.002659295	60.1	Yes	0.000210472	0.00012663
GG Maximum Velocity (Y) 12	[m/s]	7.112316721	7.100295697	6.988360337	7.423120048	100	Yes	0.049183679	0.094422126
GG Minimum Velocity (Z) 13	[m/s]	-29.4361755	-31.04460172	-32.73813224	-29.43617545	39.6	Yes	3.301956792	1.308610721
GG Average Velocity (Z) 14	[m/s]	19.76838408	19.76829392	19.76720997	19.76907194	100	Yes	0.000274475	0.005742452
GG Maximum Velocity (Z) 15	[m/s]	23.01661919	23.00493976	22.98749505	23.03063322	100	Yes	0.004133655	0.103637941
GG Force (Y) 16	[N]	0.087296409	0.068617543	0.048940077	0.087296409	100	Yes	0.007760231	0.019198839
GG Friction Force (Z) 17	[N]	0.172628622	0.175330687	0.171141405	0.179385843	100	Yes	0.000416277	0.006803728

Iterations [ ]: 310
Analysis interval: 39

Analysis Reports

Condition 1: Turning Condition



Analysis Reports

nt7.SLDASM [t7 [Default]]

Goal Name	Unit	Value	Averaged Value	Minimum Value	Maximum Value	Progress [%]	Use In Convergence	Delta	Criteria
GG Maximum Static Pressure 1	[Pa]	101335.8633	101335.8493	101335.7941	101335.9029	100	Yes	0.10886291	18.93927388
GG Maximum Total Pressure 2	[Pa]	101355.1019	101355.0752	101355.053	101355.1019	100	Yes	0.00633406	18.49530803
GG Maximum Dynamic Pressure 3	[Pa]	23.36226888	23.35226989	23.33042241	23.37433505	100	Yes	0.002624592	1.230676214
GG Maximum Total Temperature 4	[K]	303.1652865	303.165469	303.1652656	303.1656775	100	Yes	4.95967E-05	5.40634E-05
GG Maximum Velocity 5	[m/s]	6.335328798	6.33397257	6.331009293	6.336964492	100	Yes	0.000355964	0.14045339
GG Maximum Velocity (X) 6	[m/s]	1.515246473	1.528550671	1.341449899	1.646910808	100	Yes	0.128081824	0.247068211
GG Maximum Velocity (Y) 7	[m/s]	1.920389402	1.912691555	1.903412395	1.92071963	100	Yes	0.017307235	0.063709973
GG Maximum Velocity (Z) 8	[m/s]	6.315541483	6.314206685	6.311154453	6.31711036	100	Yes	0.000547989	0.031865883
GG Maximum Mach Number 9	[ ]	0.018157608	0.01815372	0.018145227	0.018162296	100	Yes	1.02029E-06	0.00036051
GG Force (Y) 10	[N]	-0.00650695	-0.006001937	-0.007145073	-0.004690316	100	Yes	0.001094884	0.00220587
GG Friction Force (Z) 11	[N]	0.021439246	0.021208959	0.020769838	0.021494642	100	Yes	5.24357E-05	0.000306646

Iterations [ ]: 144

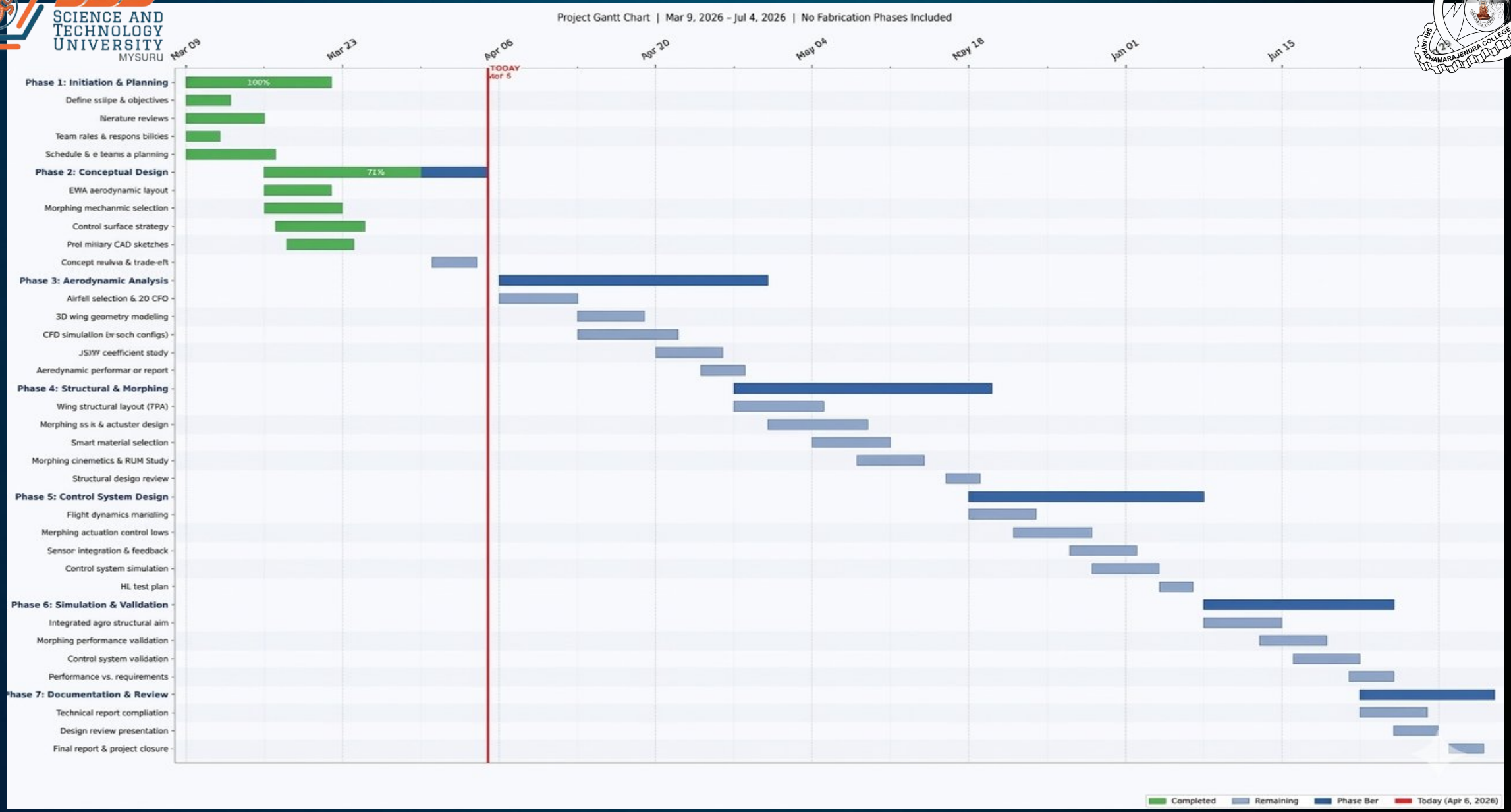
Analysis interval: 39

Applications:

- **Adaptive Lift Distribution:** Morphing wings can change their twist or curve along the span, shifting aerodynamic loads away from the wingtips to reduce the stress (bending moments) at the wing root.
- **Optimized Fuel Consumption:** Traditional aircraft are only perfectly efficient at one specific "cruise" speed. Morphing allows the wing to maintain its peak (Lift-to-Drag) ratio at all altitudes and speeds, resulting in documented fuel savings between 3% and 8%.
- **Stall Delay & Prevention:** Continuous camber adjustments keep the airflow attached to the wing longer at high angles of attack, effectively delaying dynamic stall.
- **Variable Flight Regimes:** By adjusting wing camber in real time, a drone can switch from high-speed dashing (low drag) to slow, stable hovering (high lift).
- **Gust Alleviation & Stability:** Just like real birds navigating tight spaces or high winds, wing morphing allows localized, rapid adjustments to reject wind gusts and maintain smooth flight.
- **Stealth & Aerodynamic Quietness:** Traditional flaps create sharp mechanical edges and gaps that reflect radar and generate noise. Continuous, seamless morphing creates a clean profile perfect for stealth reconnaissance.

Results that we can Achieve by the project:

- Energy Efficiency
- Stability
- Take-off
- standing take-off using propeller alone
- Turning
- Bio-Comparison


Project Gantt Chart | Mar 9, 2026 – Jul 4, 2026 | No Fabrication Phases Included
Mar 09
Mar 23
Apr 06
Apr 20
May 04
May 18
Jun 01
Jun 15
TODAY
Mar 5
Phase 1: Initiation & Planning
100%
Define scope & objectives
Literature reviews
Team roles & responsibilities
Schedule & milestone planning
Phase 2: Conceptual Design
71%
EWA aerodynamic layout
Morphing mechanism selection
Control surface strategy
Preliminary CAD sketches
Concept review & trade-off
Phase 3: Aerodynamic Analysis
Airfoil selection & 2D CFD
3D wing geometry modeling
CFD simulation (3 mach configs)
Lift coefficient study
Aerodynamic performance report
Phase 4: Structural & Morphing
Wing structural layout (TPA)
Morphing mechanism & actuator design
Smart material selection
Morphing kinematics & ROM study
Structural design review
Phase 5: Control System Design
Flight dynamics modelling
Morphing actuation control laws
Sensor integration & feedback
Control system simulation
HIL test plan
Phase 6: Simulation & Validation
Integrated aero structural sim
Morphing performance validation
Control system validation
Performance vs. requirements
Phase 7: Documentation & Review
Technical report compilation
Design review presentation
Final report & project closure
Completed
Remaining
Phase Bar
Today (Apr 6, 2026)